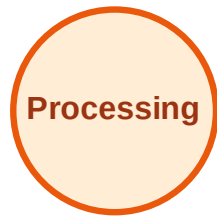


# BFS Traversal

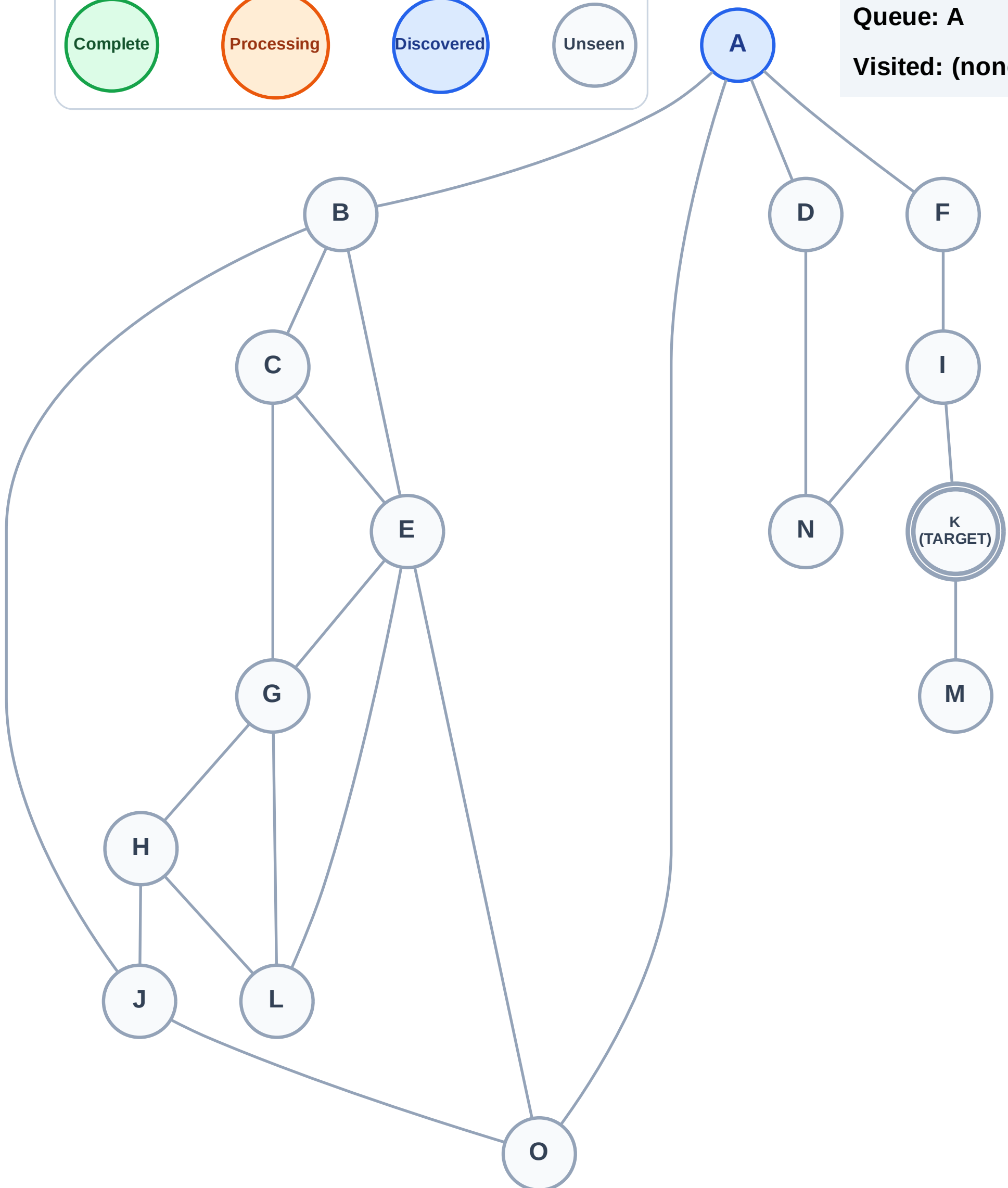
Step 1: Start at A (Seeking target node K)

## Legend



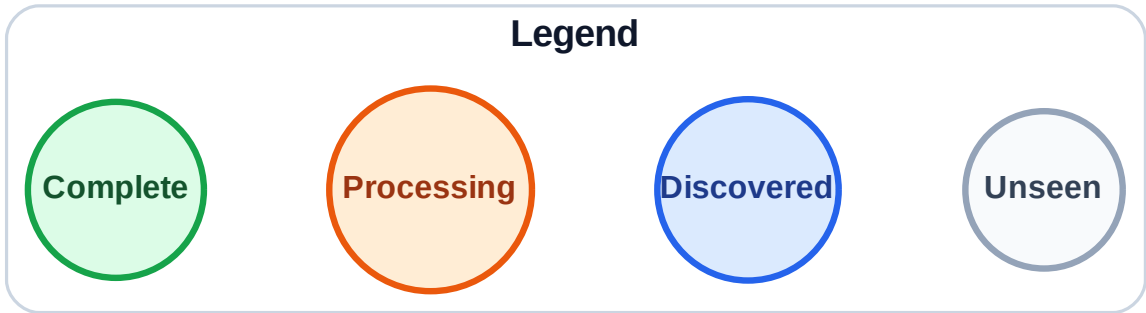
Queue: A

Visited: (none)

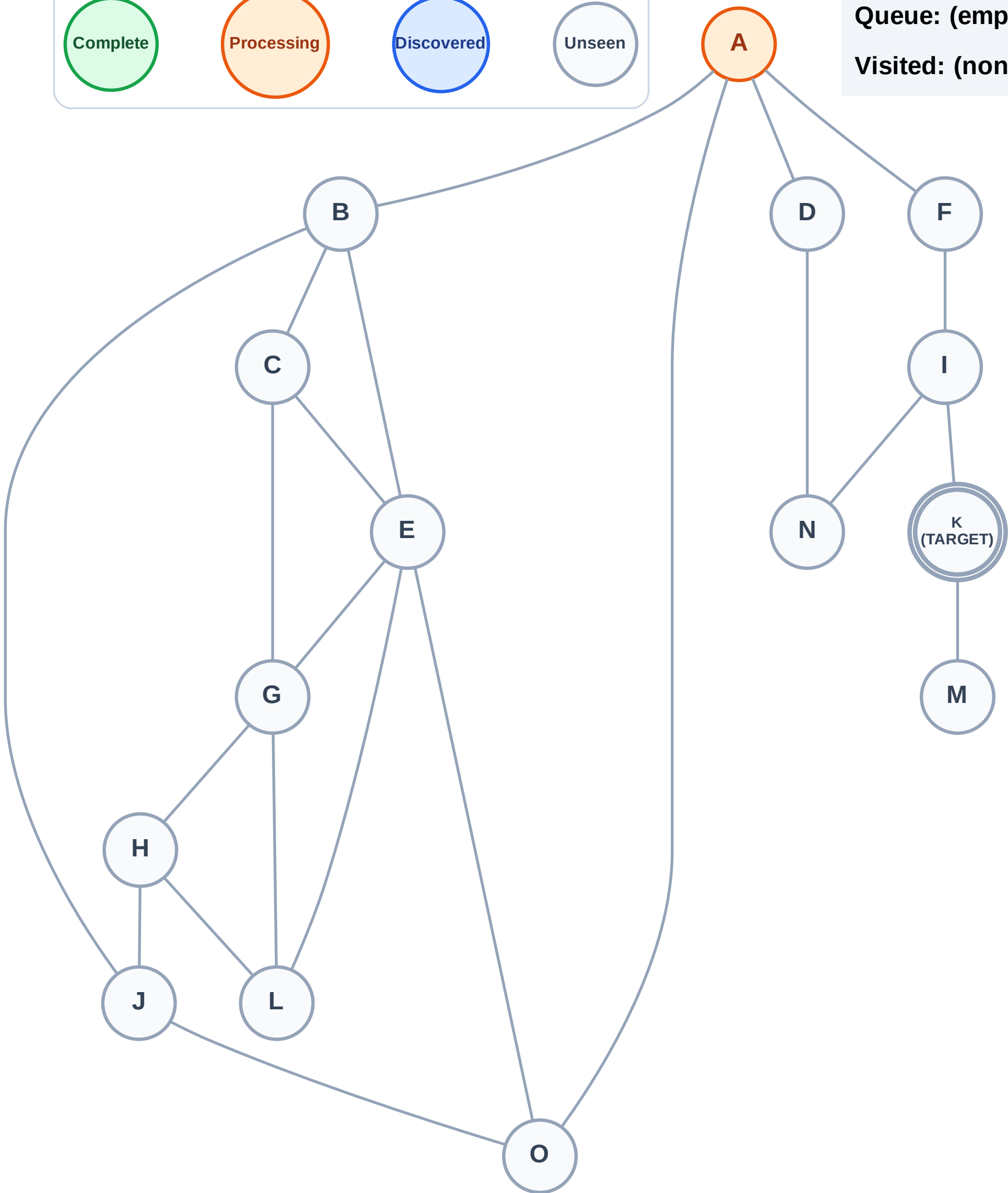


# BFS Traversal

Step 2: Process node A



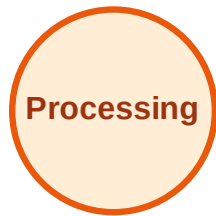
Queue: (empty)  
Visited: (none)



# BFS Traversal

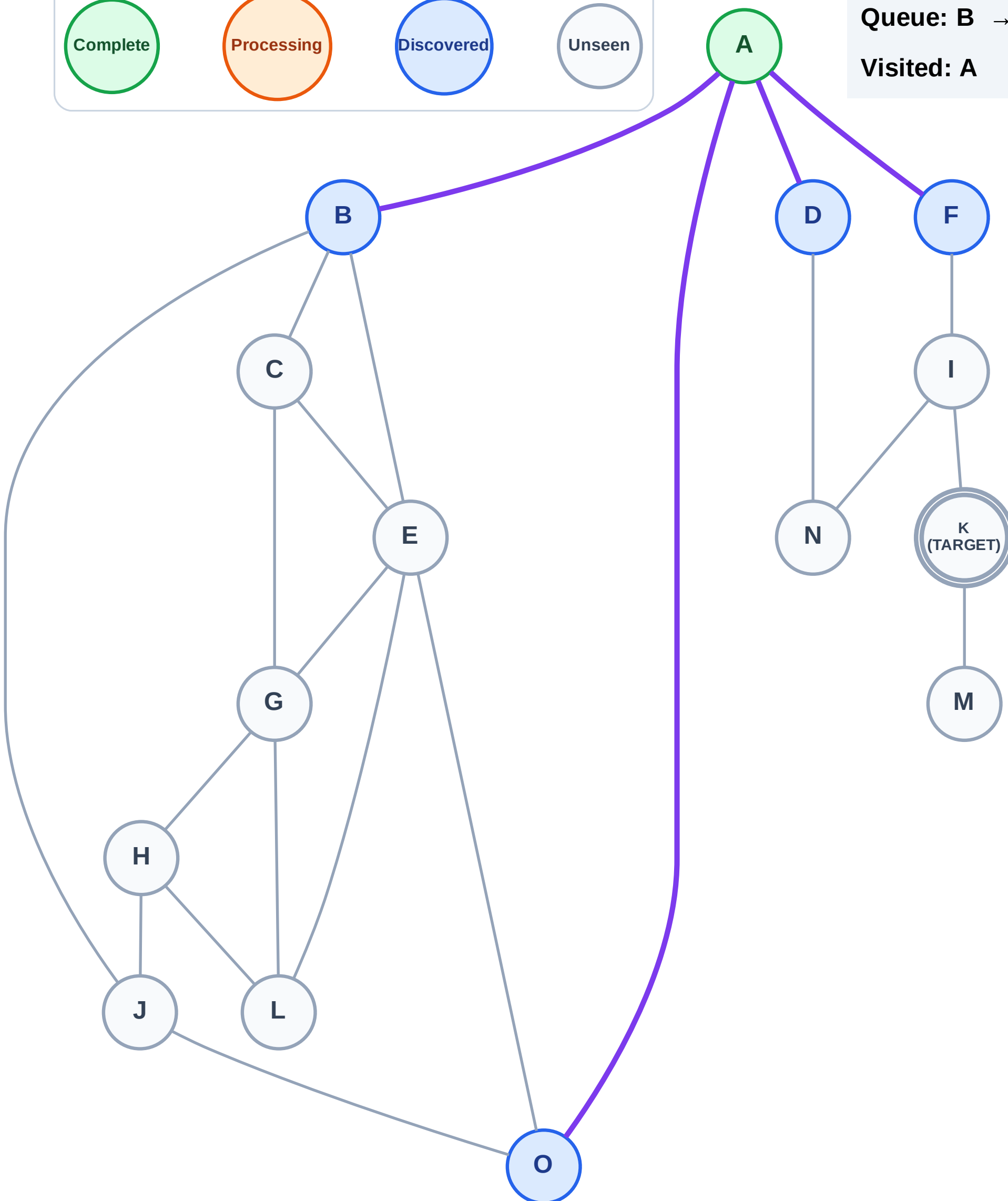
Step 3: Discover B, D, F, O from A

## Legend



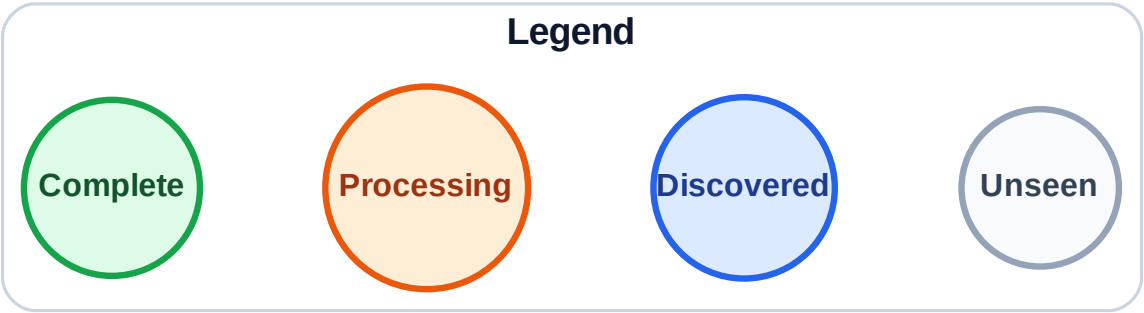
Queue: B → D → F → O

Visited: A

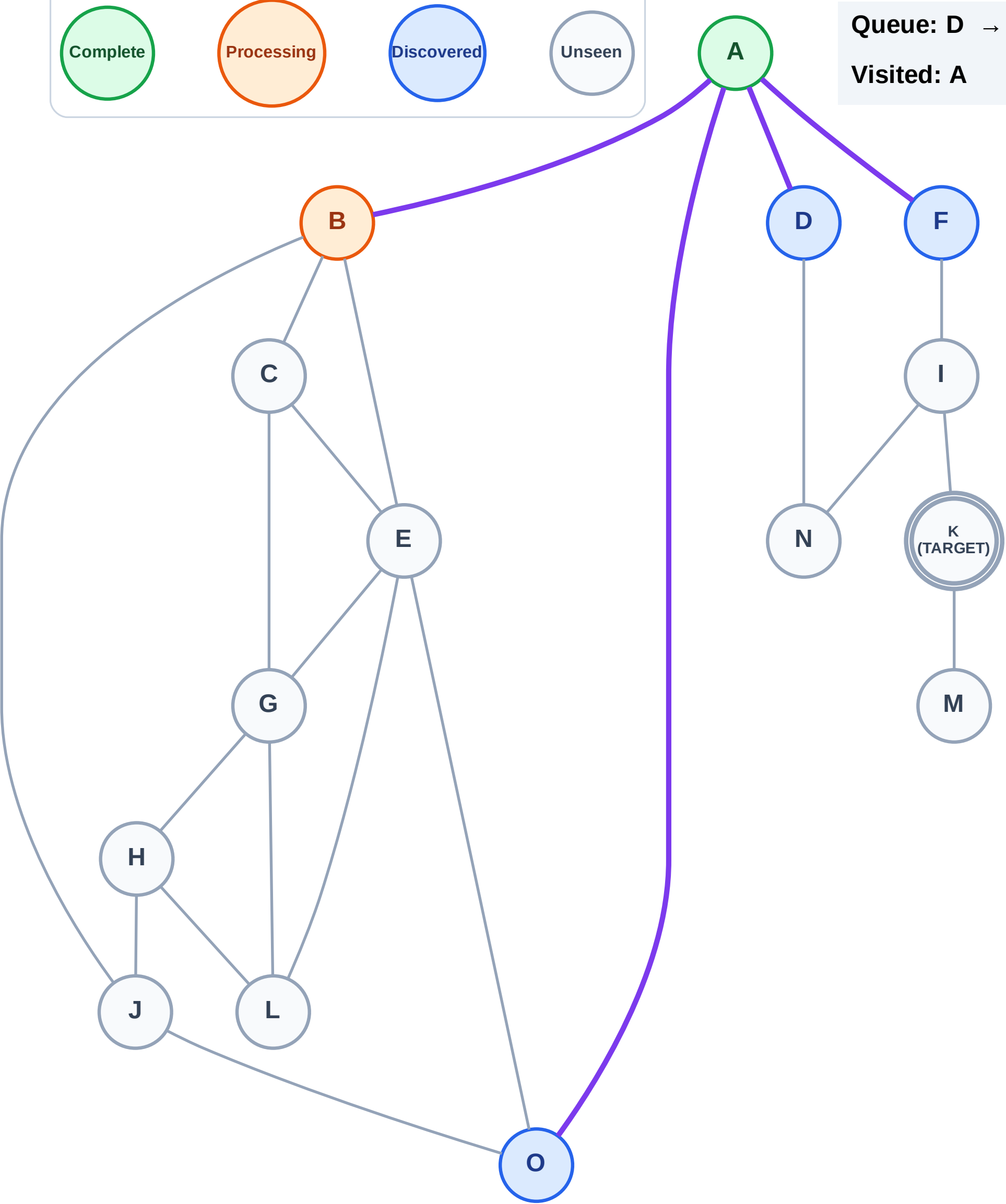


# BFS Traversal

Step 4: Process node B



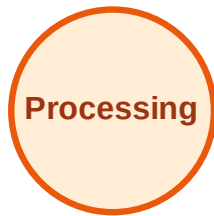
Queue: D → F → O  
Visited: A



# BFS Traversal

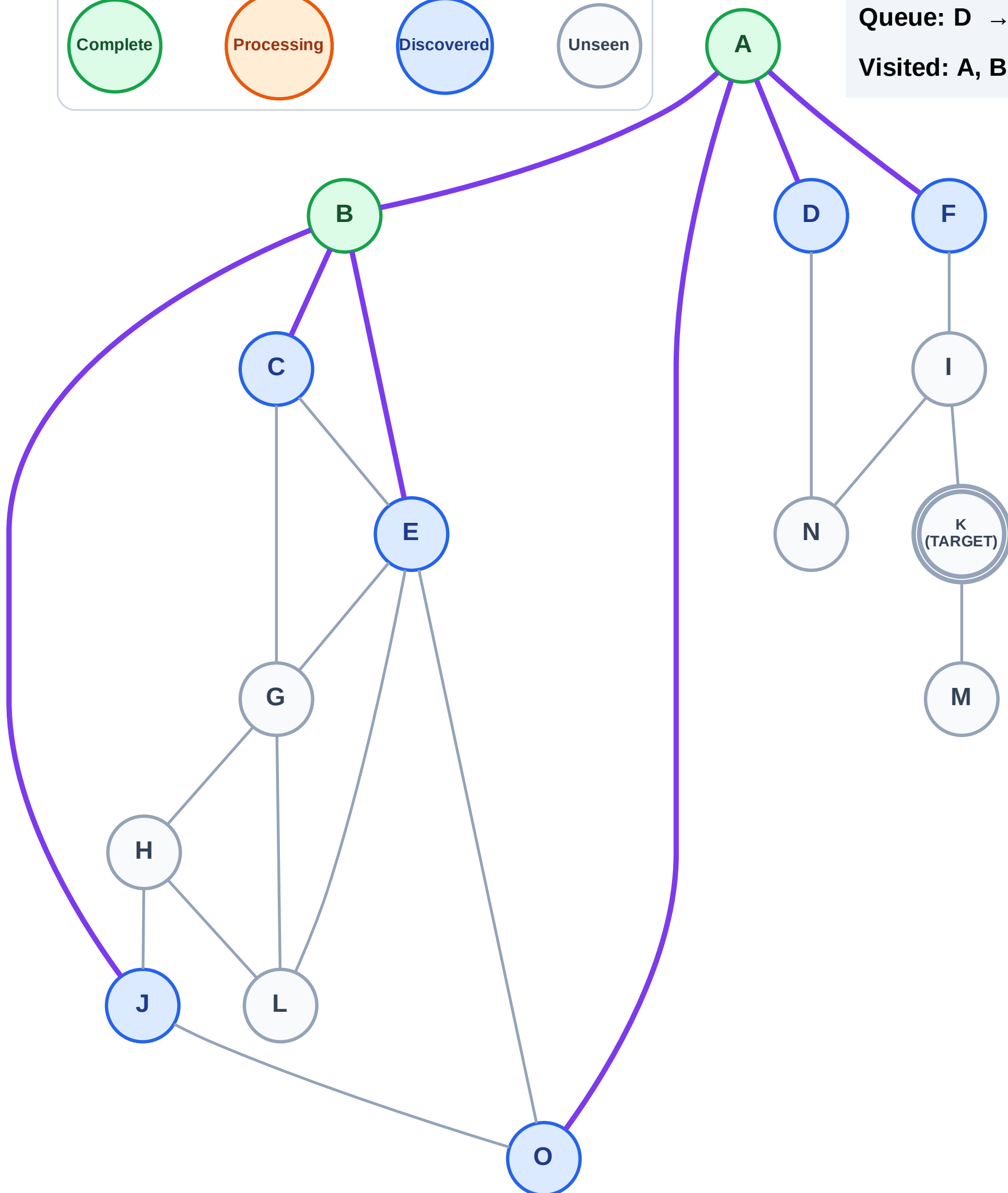
Step 5: Discover C, E, J from B

## Legend



Queue: D → F → O → C → E → J

Visited: A, B



# BFS Traversal

Step 6: Process node D

Legend

Complete

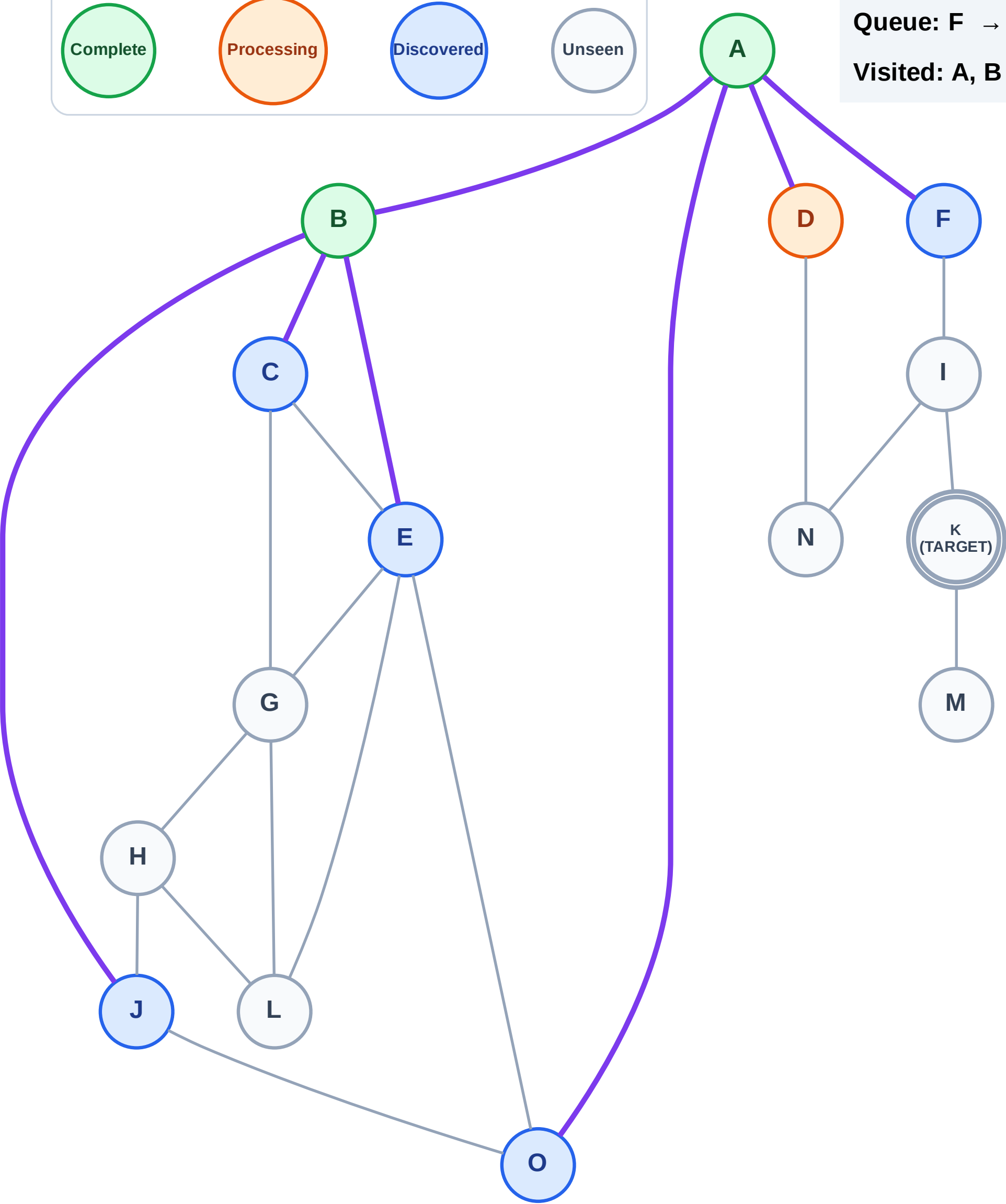
Processing

Discovered

Unseen

Queue: F → O → C → E → J

Visited: A, B



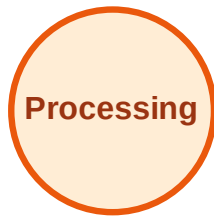
# BFS Traversal

Step 7: Discover N from D

## Legend



Complete



Processing



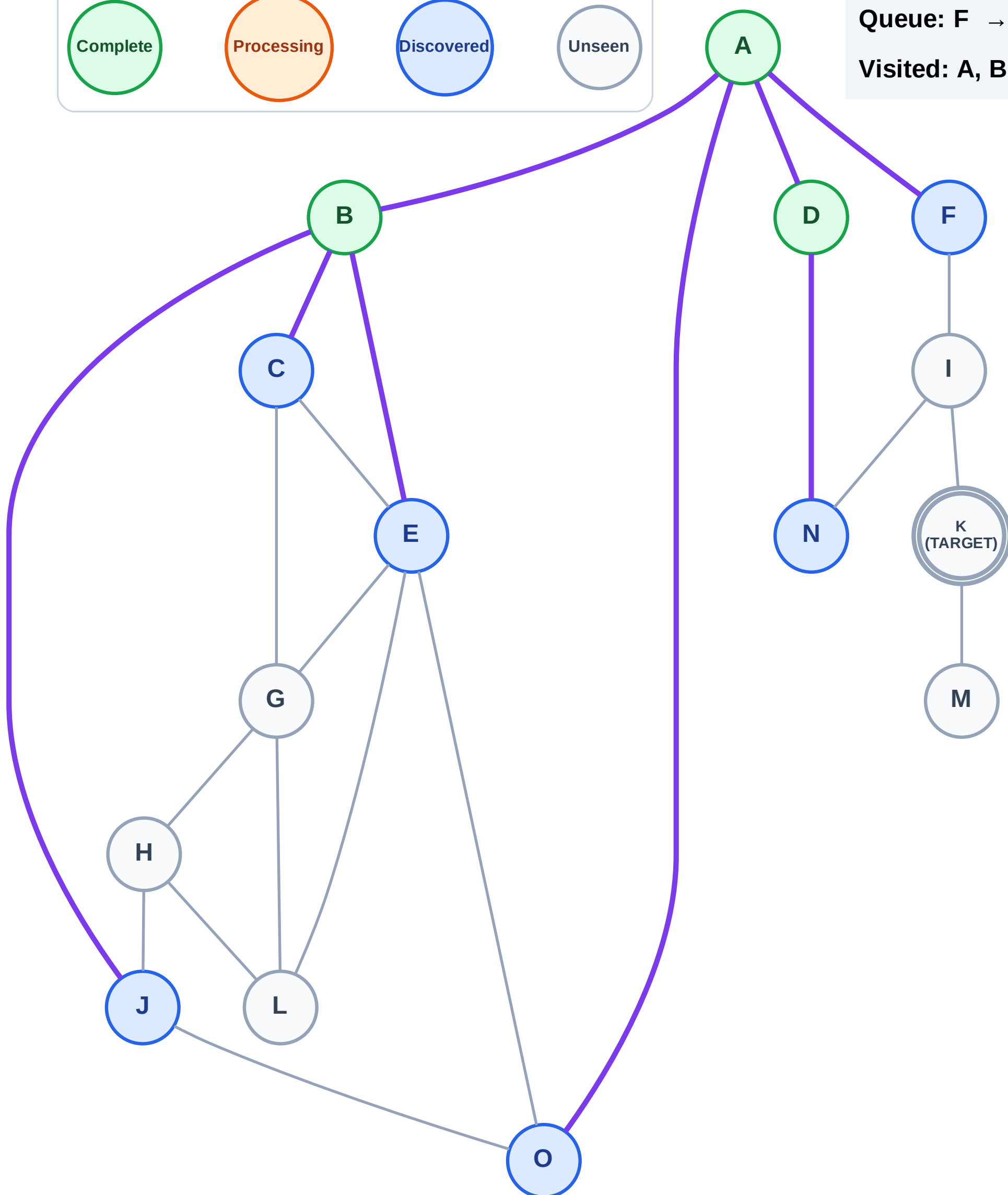
Discovered



Unseen

Queue: F → O → C → E → J → N

Visited: A, B, D



# BFS Traversal

Step 8: Process node F

Legend

Complete

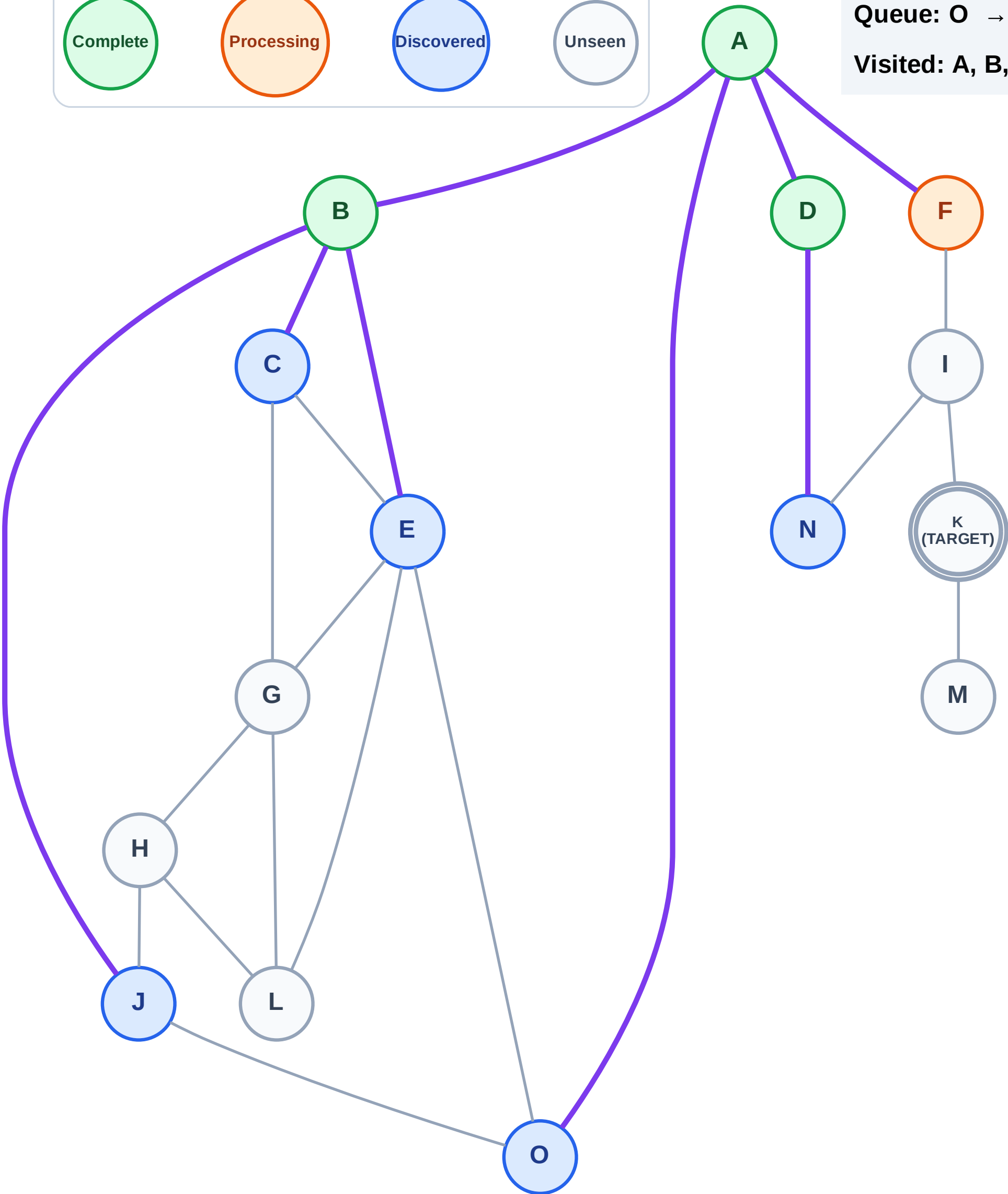
Processing

Discovered

Unseen

Queue: O → C → E → J → N

Visited: A, B, D



# BFS Traversal

Step 9: Discover I from F

Legend

Complete

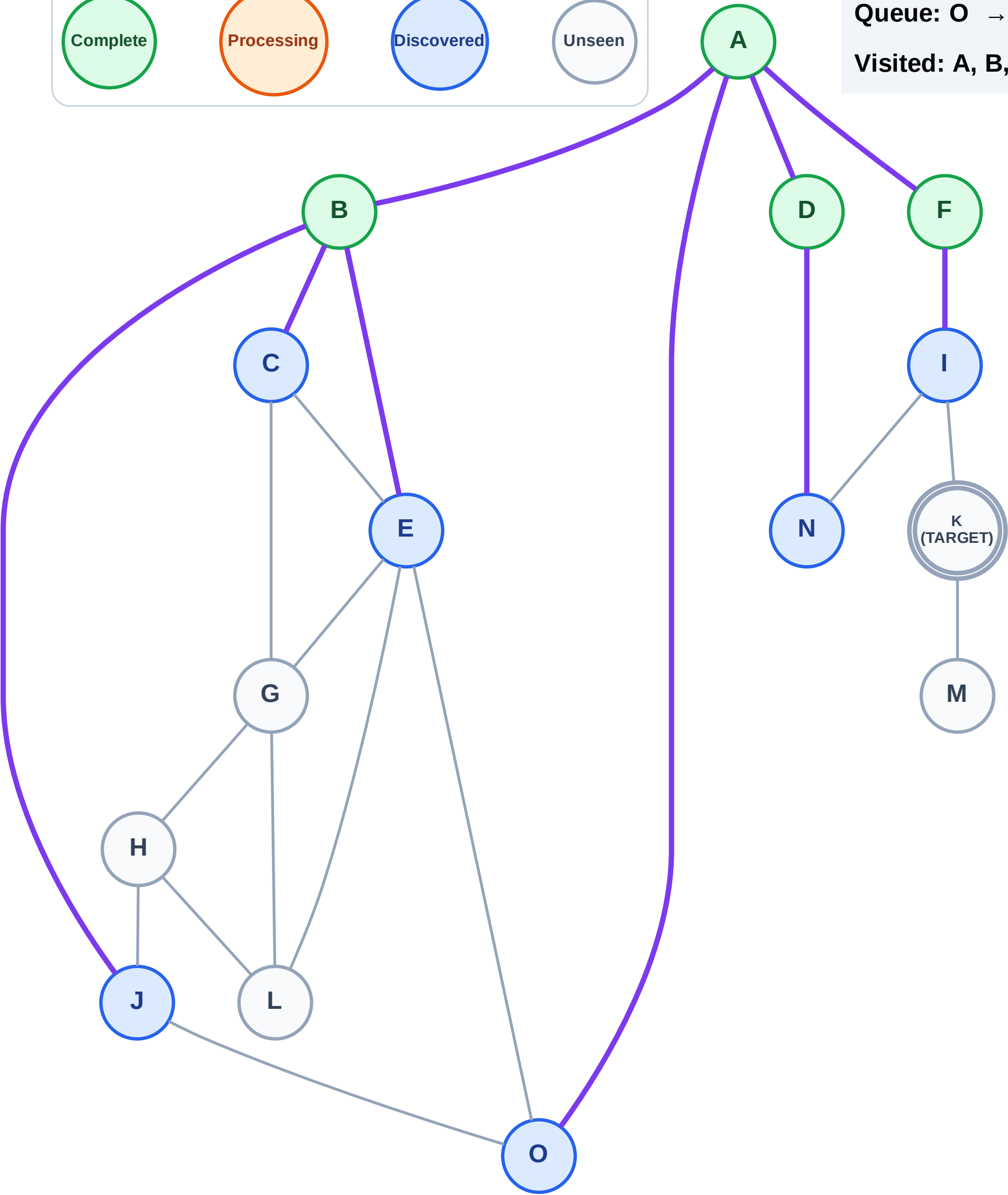
Processing

Discovered

Unseen

Queue: O → C → E → J → N → I

Visited: A, B, D, F



# BFS Traversal

Step 10: Process node O

## Legend



Complete



Processing



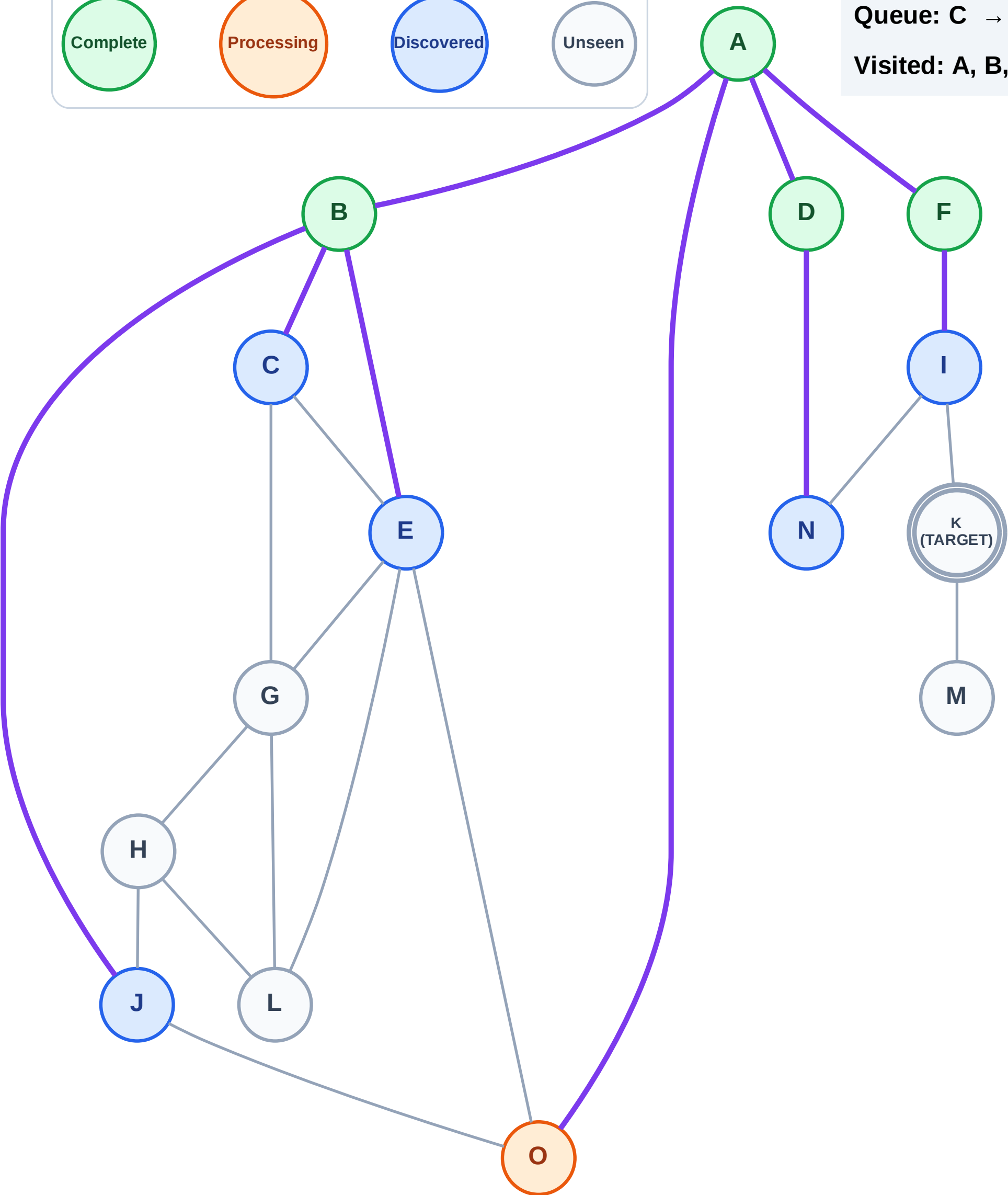
Discovered



Unseen

Queue: C → E → J → N → I

Visited: A, B, D, F



# BFS Traversal

Step 11: No new neighbors from O

## Legend



Complete



Processing



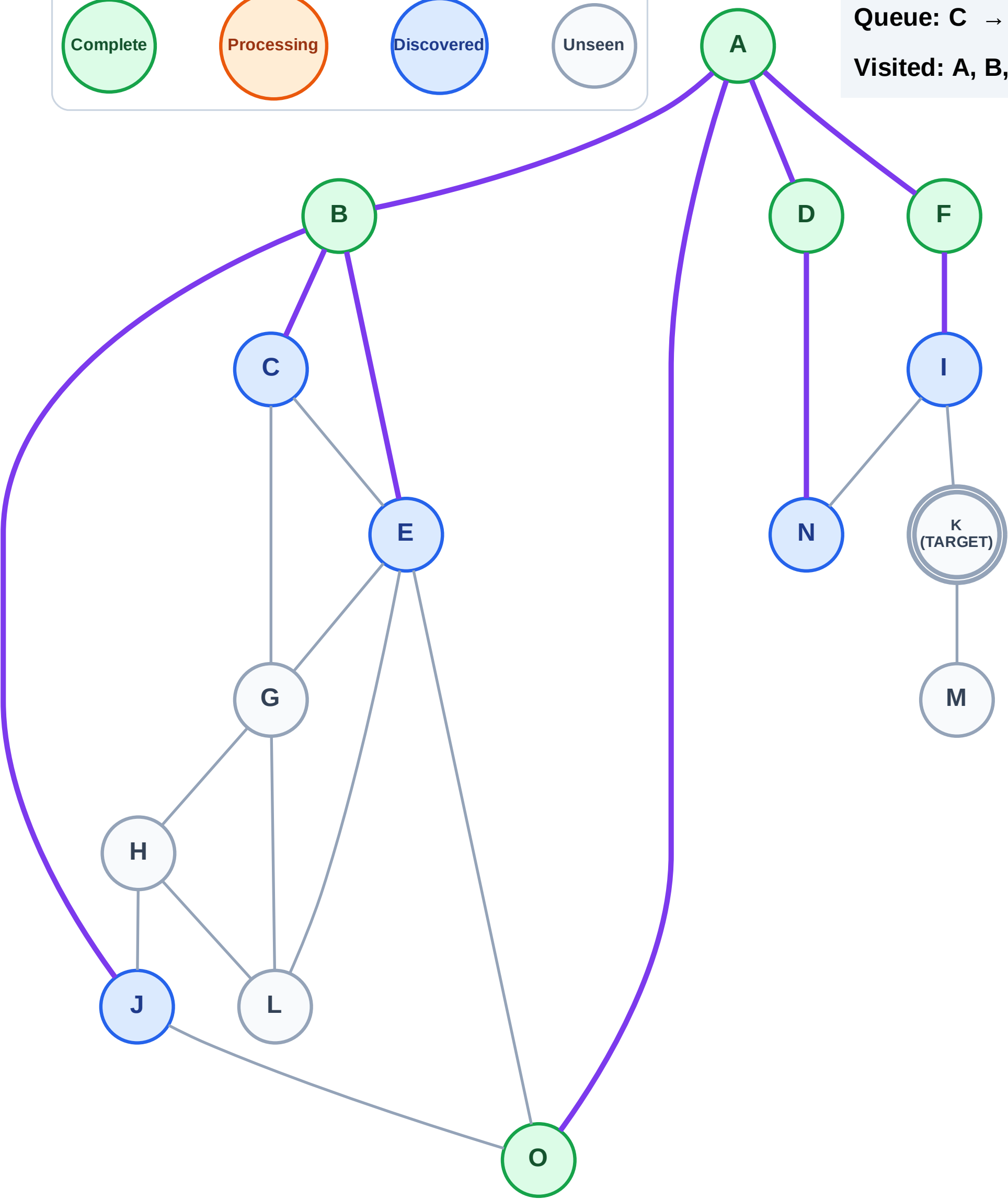
Discovered



Unseen

Queue: C → E → J → N → I

Visited: A, B, D, F, O



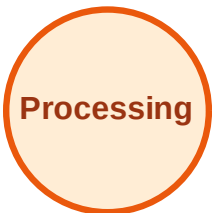
# BFS Traversal

Step 12: Process node C

## Legend



Complete



Processing



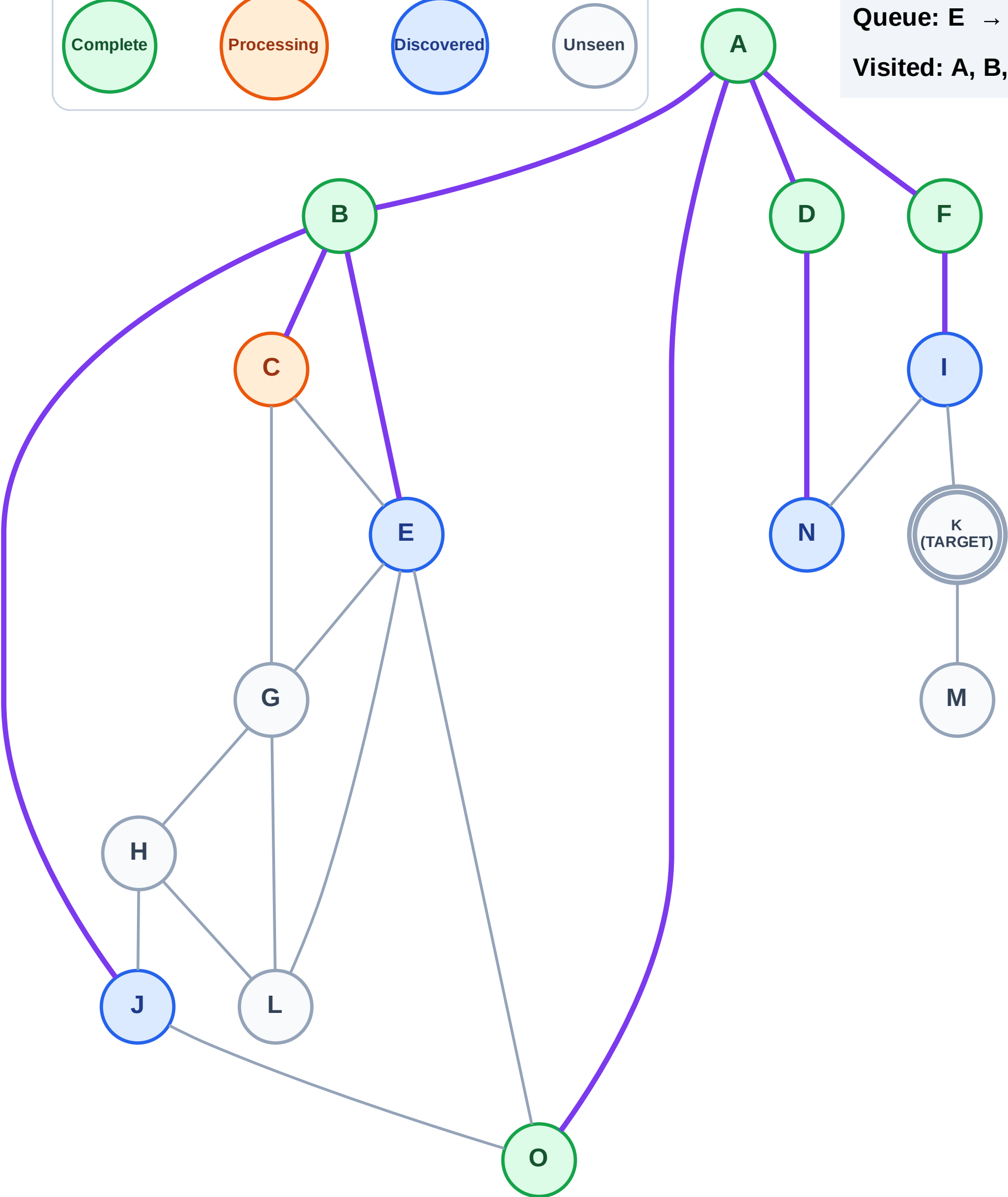
Discovered



Unseen

Queue: E → J → N → I

Visited: A, B, D, F, O



# BFS Traversal

Step 13: Discover G from C

## Legend



Complete



Processing



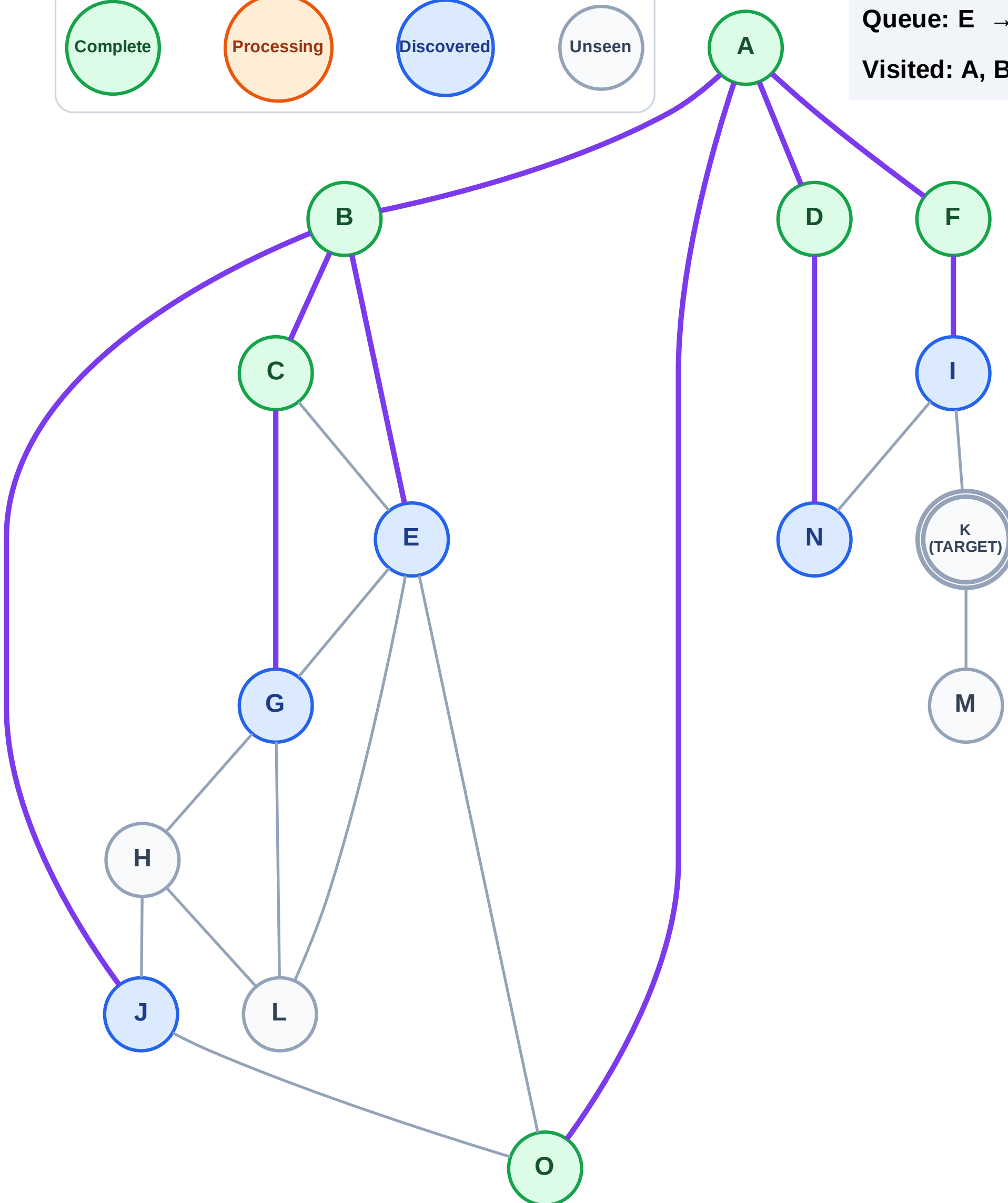
Discovered



Unseen

Queue: E → J → N → I → G

Visited: A, B, C, D, F, O



# BFS Traversal

Step 14: Process node E

Legend

Complete

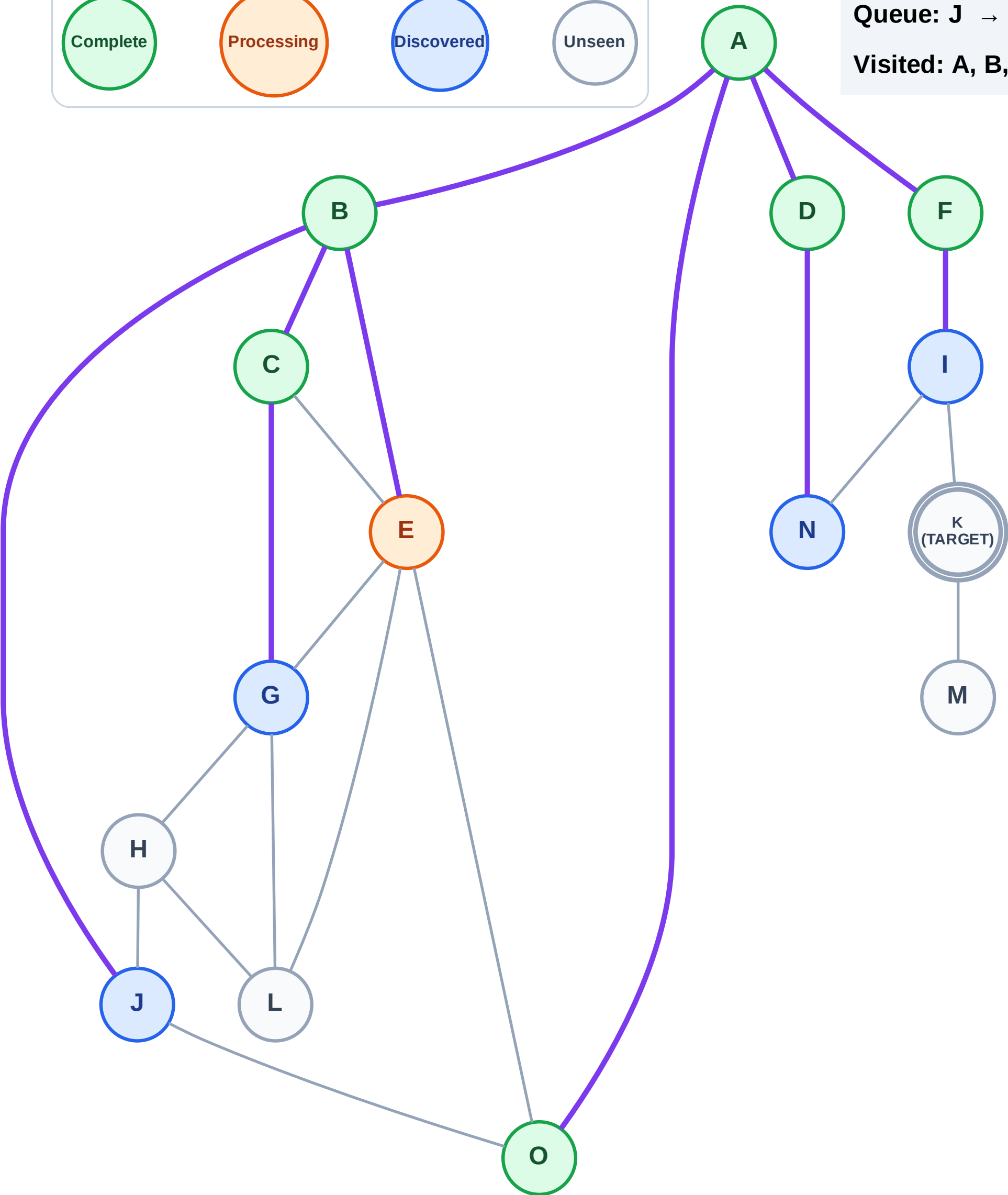
Processing

Discovered

Unseen

Queue: J → N → I → G

Visited: A, B, C, D, F, O



# BFS Traversal

Step 15: Discover L from E

Legend

Complete

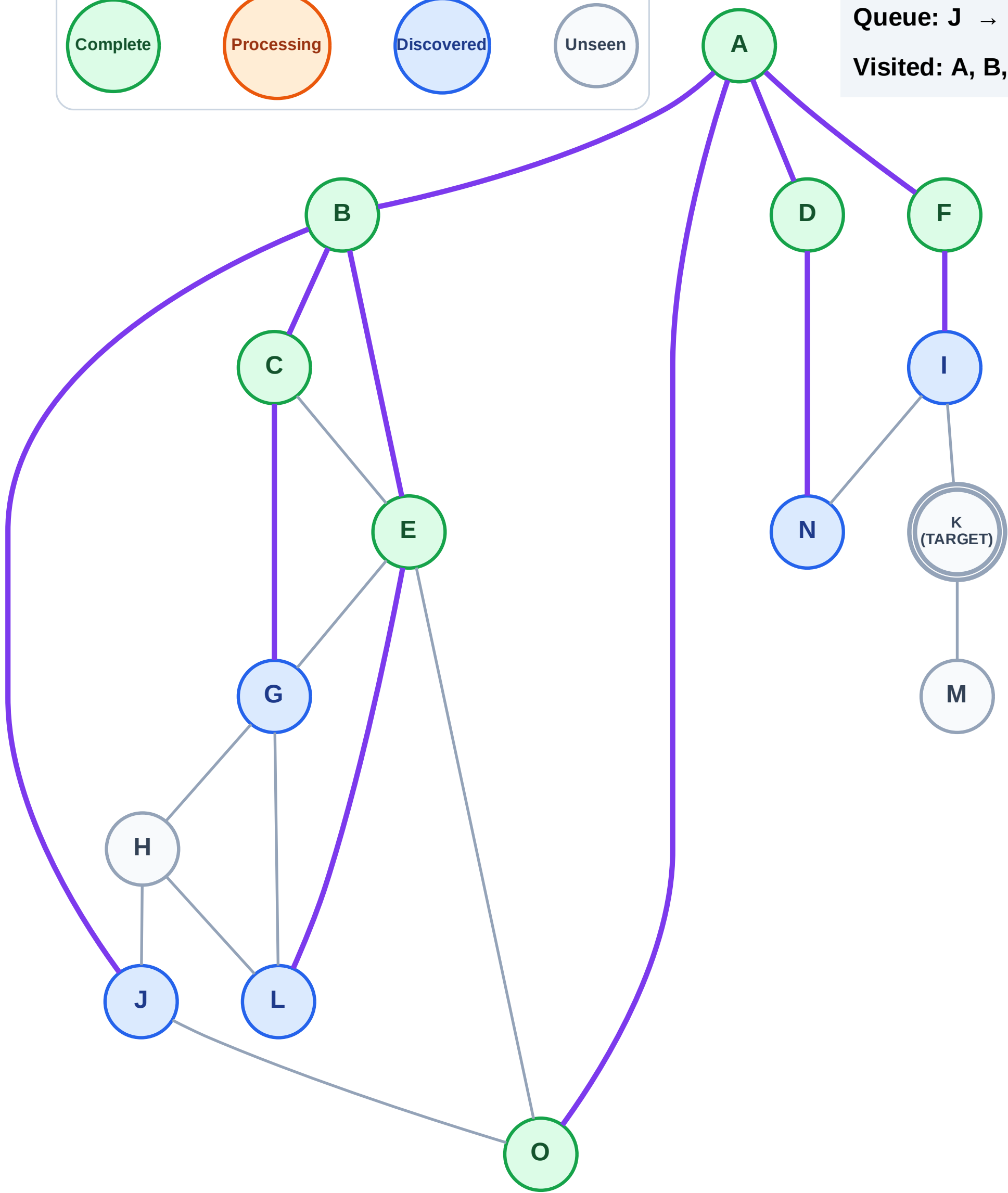
Processing

Discovered

Unseen

Queue: J → N → I → G → L

Visited: A, B, C, D, E, F, O



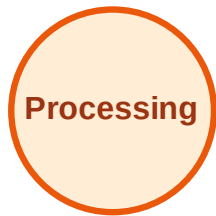
# BFS Traversal

Step 16: Process node J

## Legend



Complete



Processing



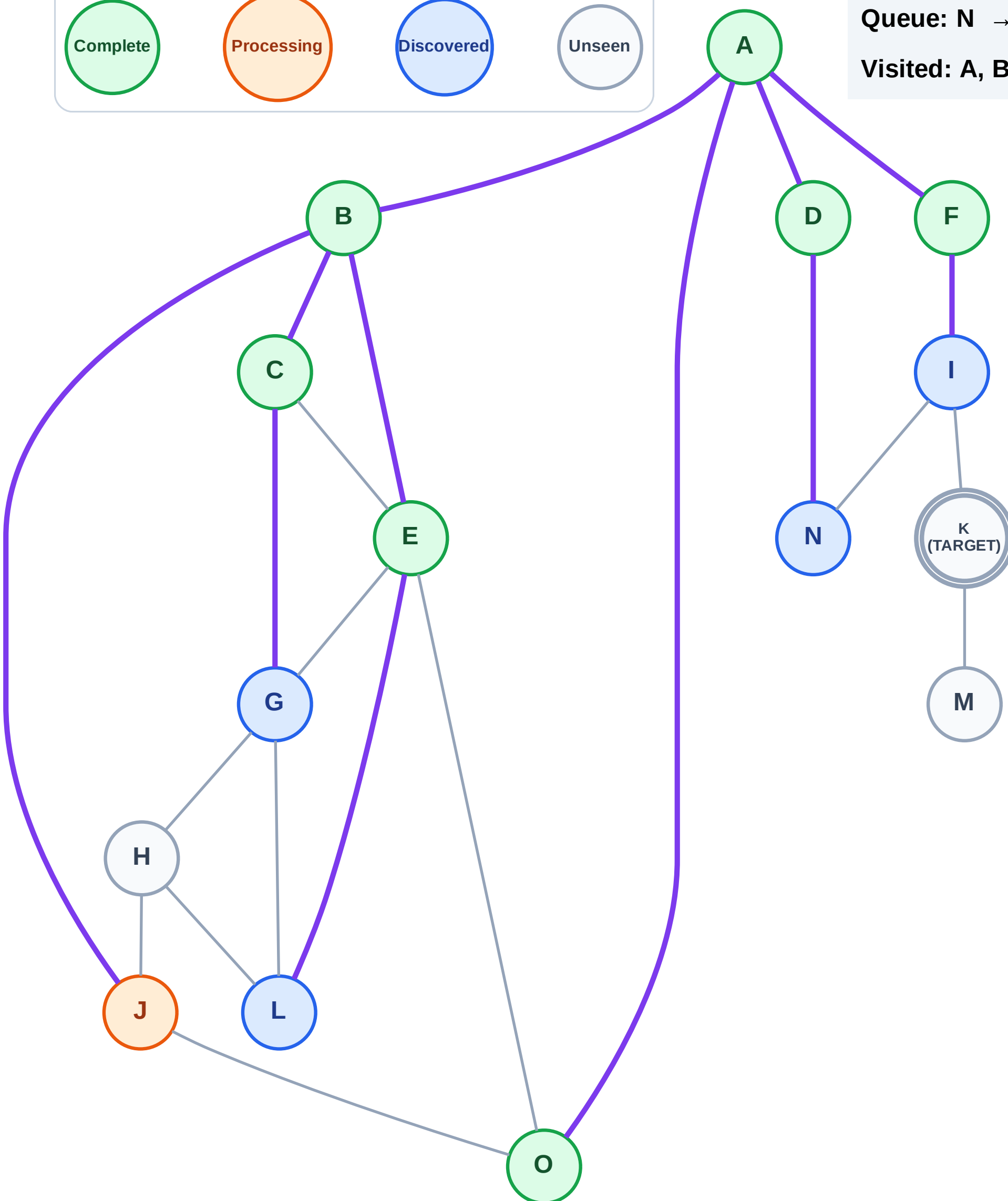
Discovered



Unseen

Queue: N → I → G → L

Visited: A, B, C, D, E, F, O



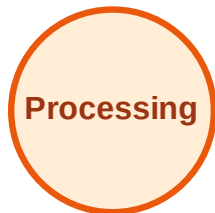
# BFS Traversal

Step 17: Discover H from J

## Legend



Complete



Processing



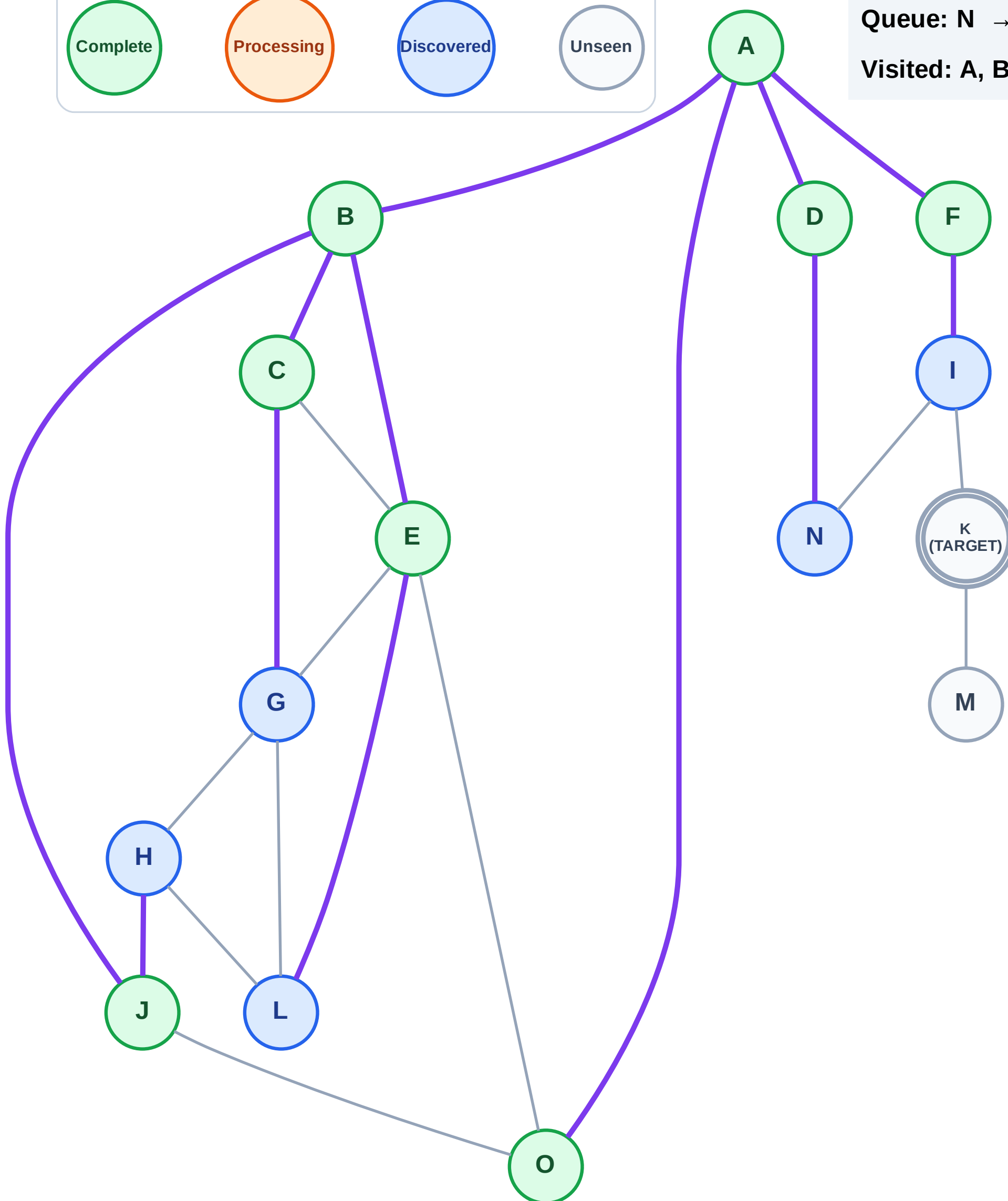
Discovered



Unseen

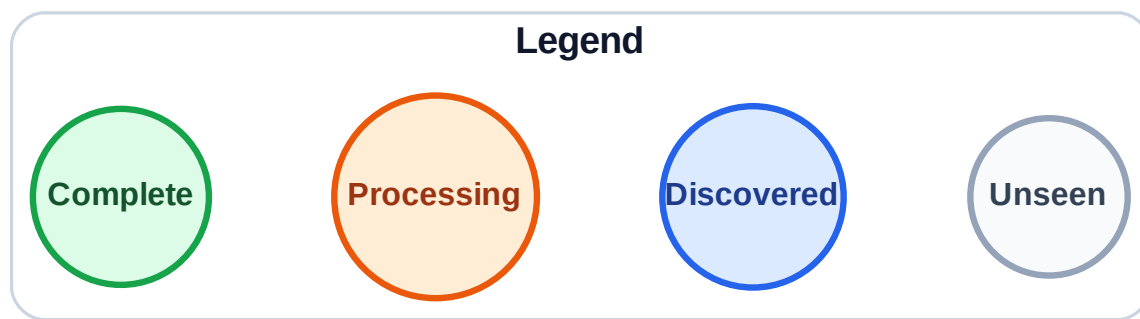
Queue: N → I → G → L → H

Visited: A, B, C, D, E, F, J, O

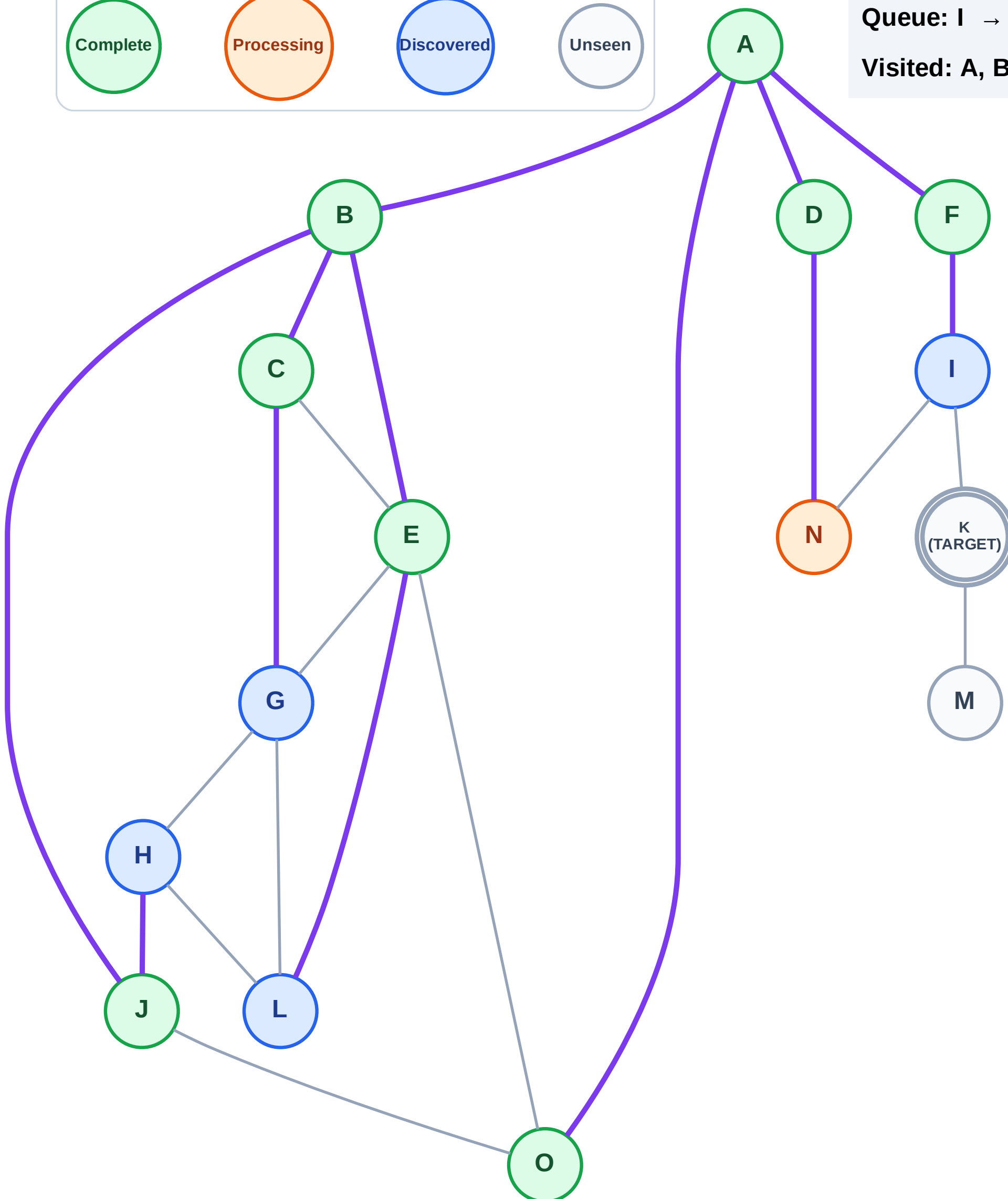


# BFS Traversal

Step 18: Process node N



Queue: I → G → L → H  
Visited: A, B, C, D, E, F, J, O



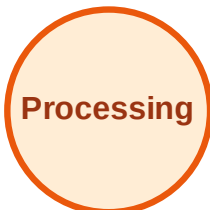
# BFS Traversal

Step 19: No new neighbors from N

## Legend



Complete



Processing



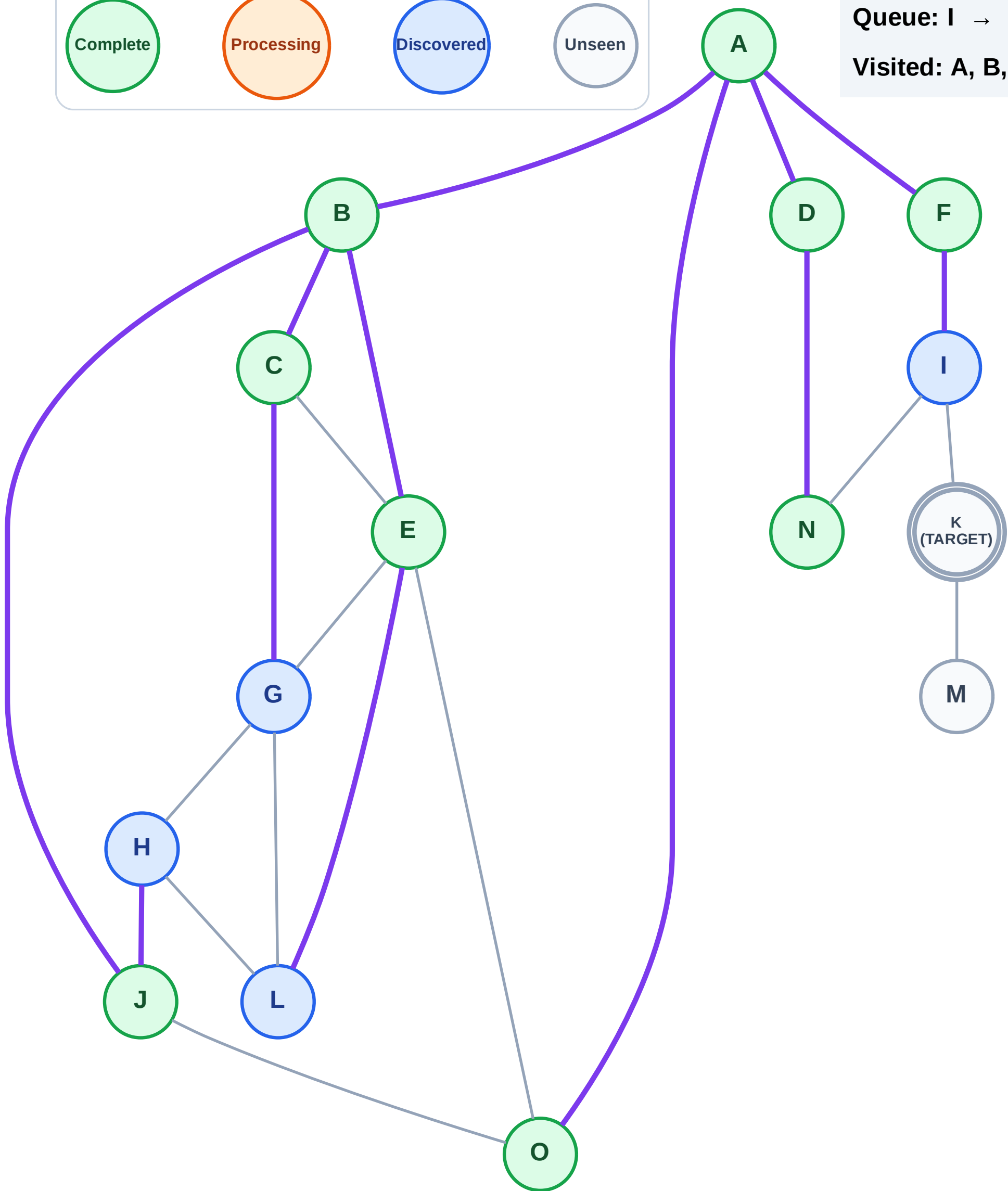
Discovered



Unseen

Queue: I → G → L → H

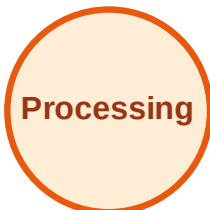
Visited: A, B, C, D, E, F, J, N, O



# BFS Traversal

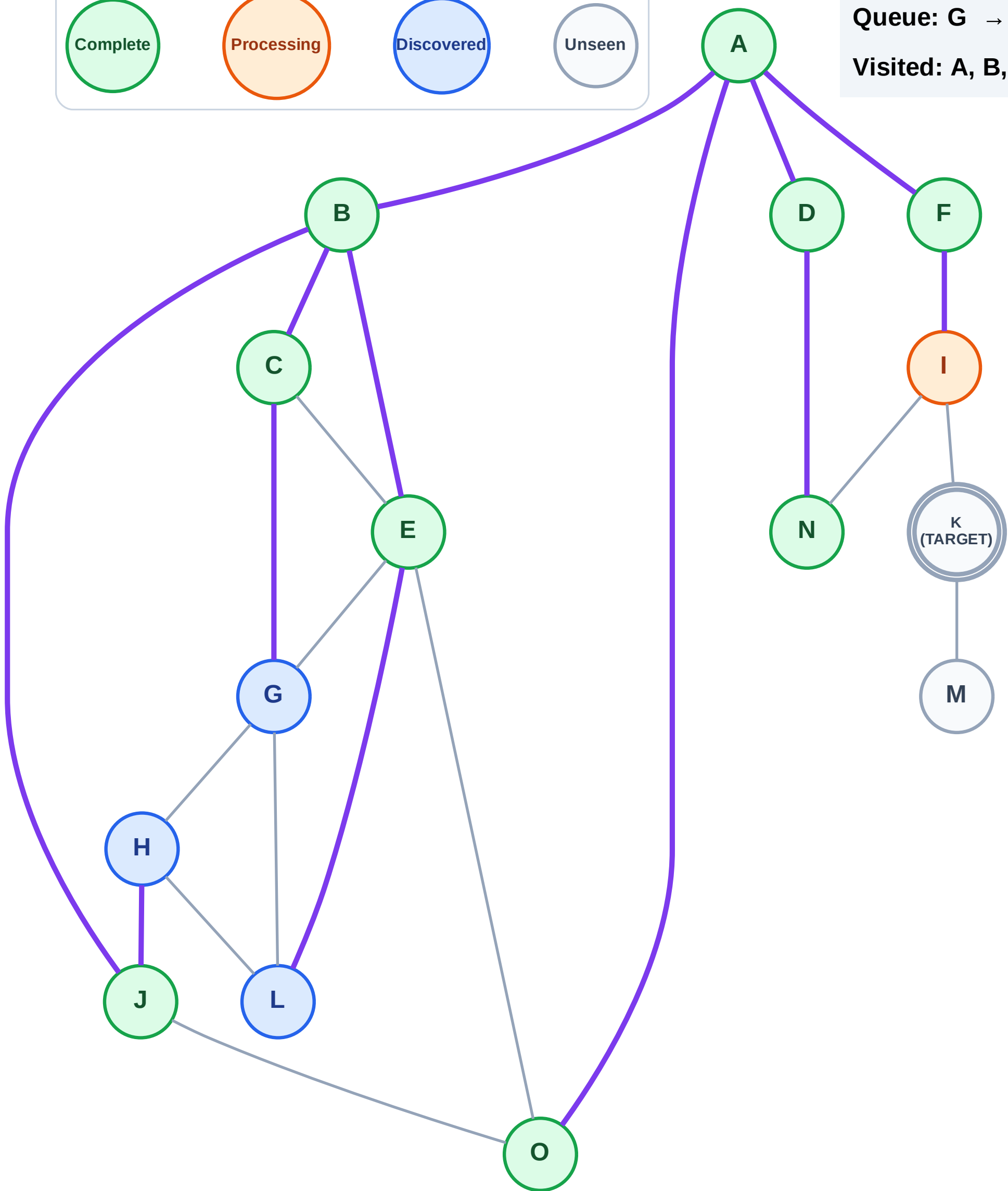
Step 20: Process node I

## Legend



Queue: G → L → H

Visited: A, B, C, D, E, F, J, N, O



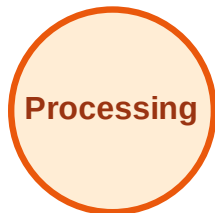
# BFS Traversal

Step 21: Discover K from I

## Legend



Complete



Processing



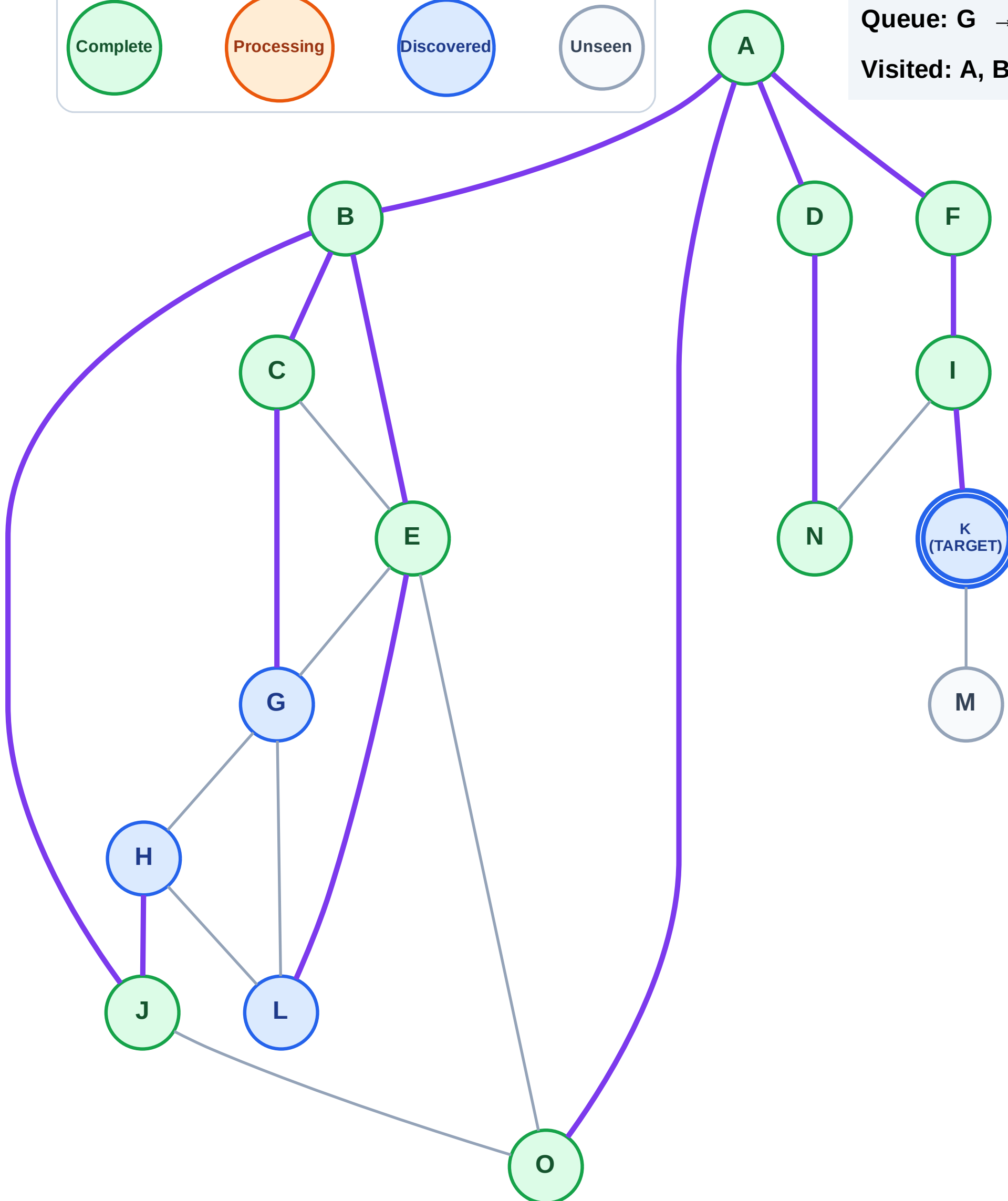
Discovered



Unseen

Queue: G → L → H → K

Visited: A, B, C, D, E, F, I, J, N, O



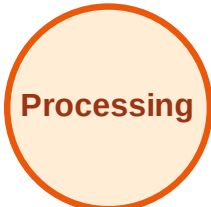
# BFS Traversal

Step 22: Process node G

## Legend



Complete



Processing



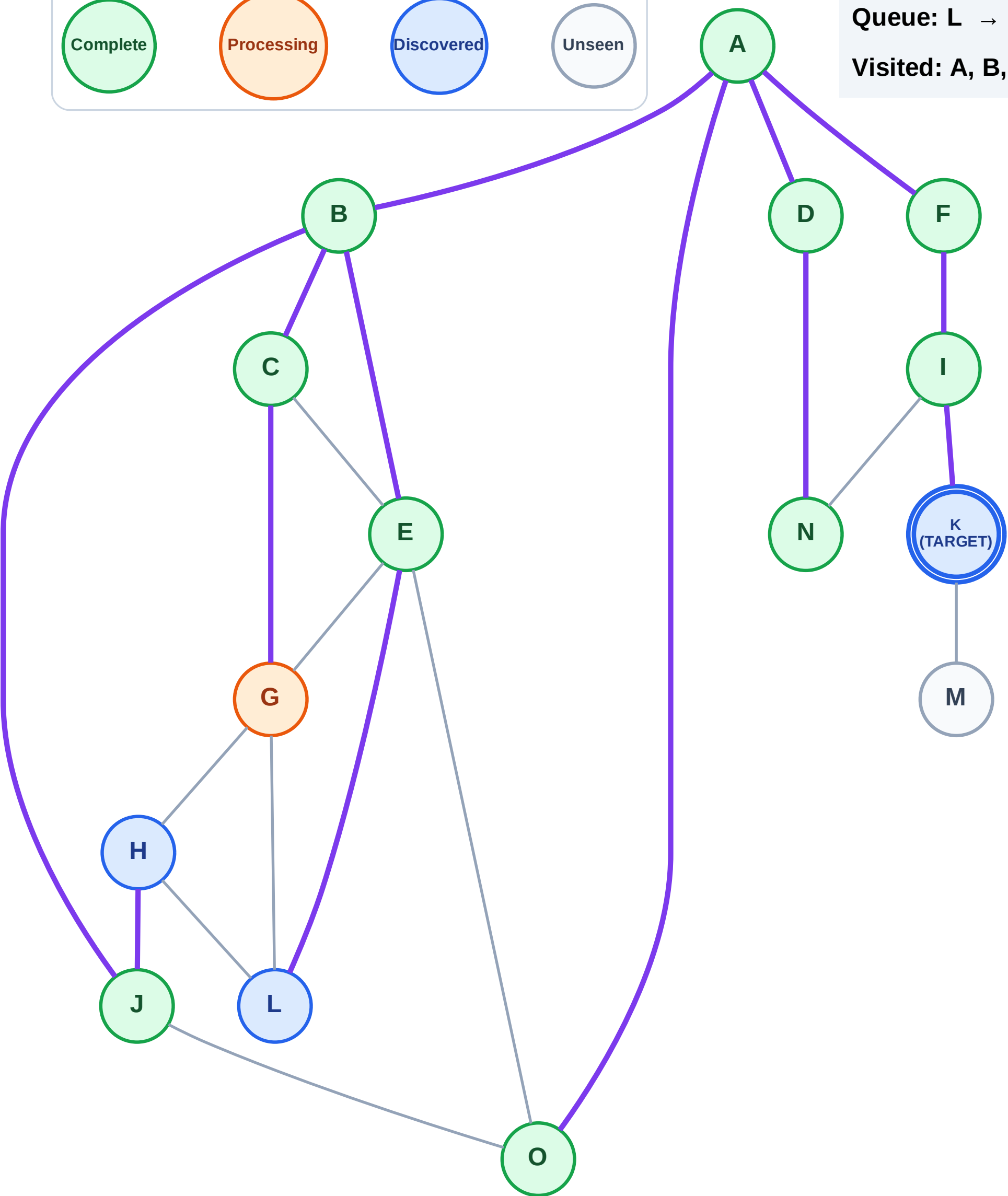
Discovered



Unseen

Queue: L → H → K

Visited: A, B, C, D, E, F, I, J, N, O



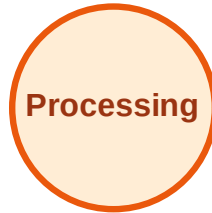
# BFS Traversal

Step 23: No new neighbors from G

## Legend



Complete



Processing



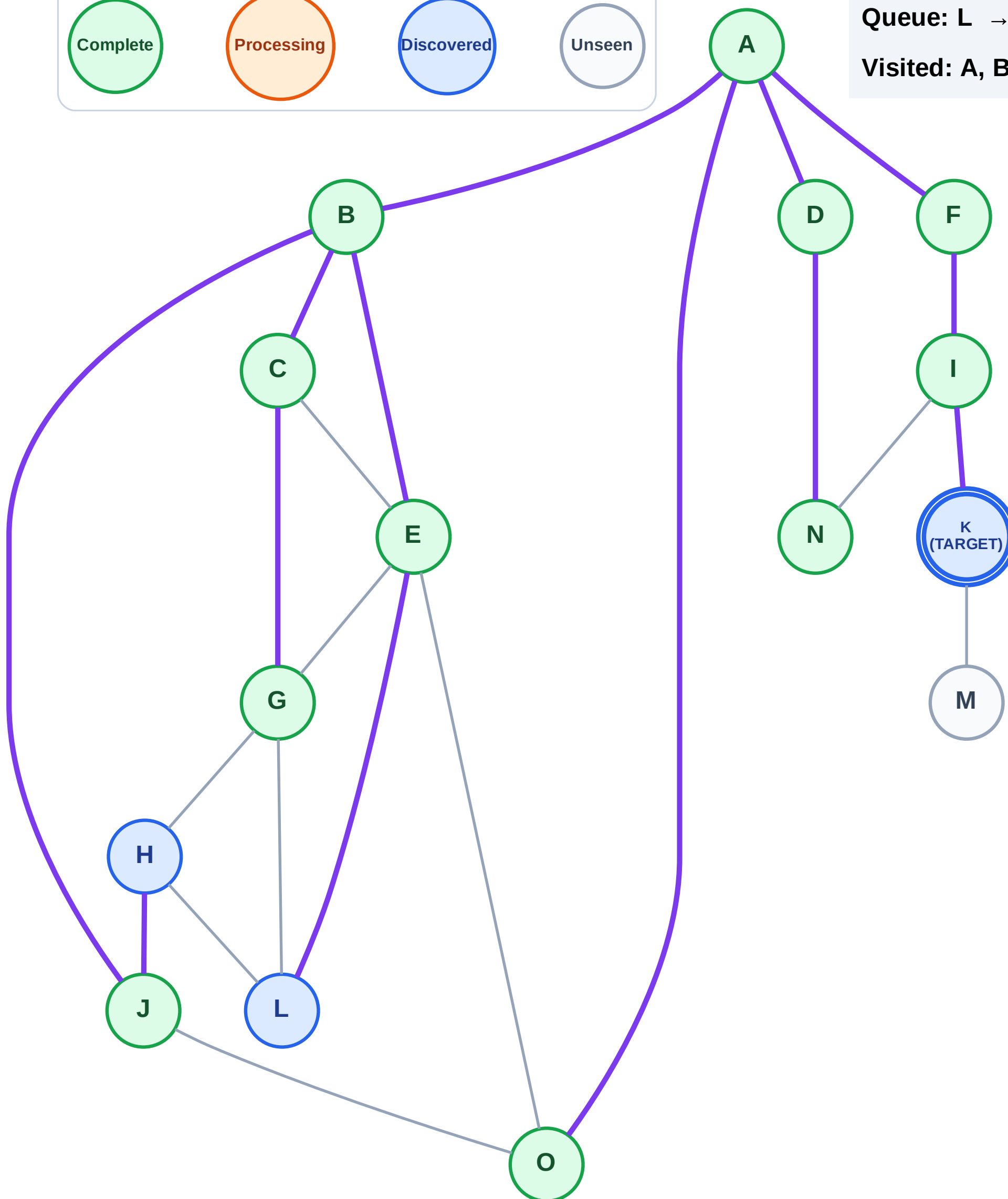
Discovered



Unseen

Queue: L → H → K

Visited: A, B, C, D, E, F, G, I, J, N, O



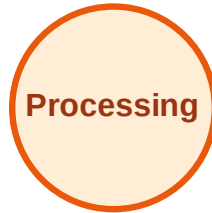
# BFS Traversal

Step 24: Process node L

## Legend



Complete



Processing



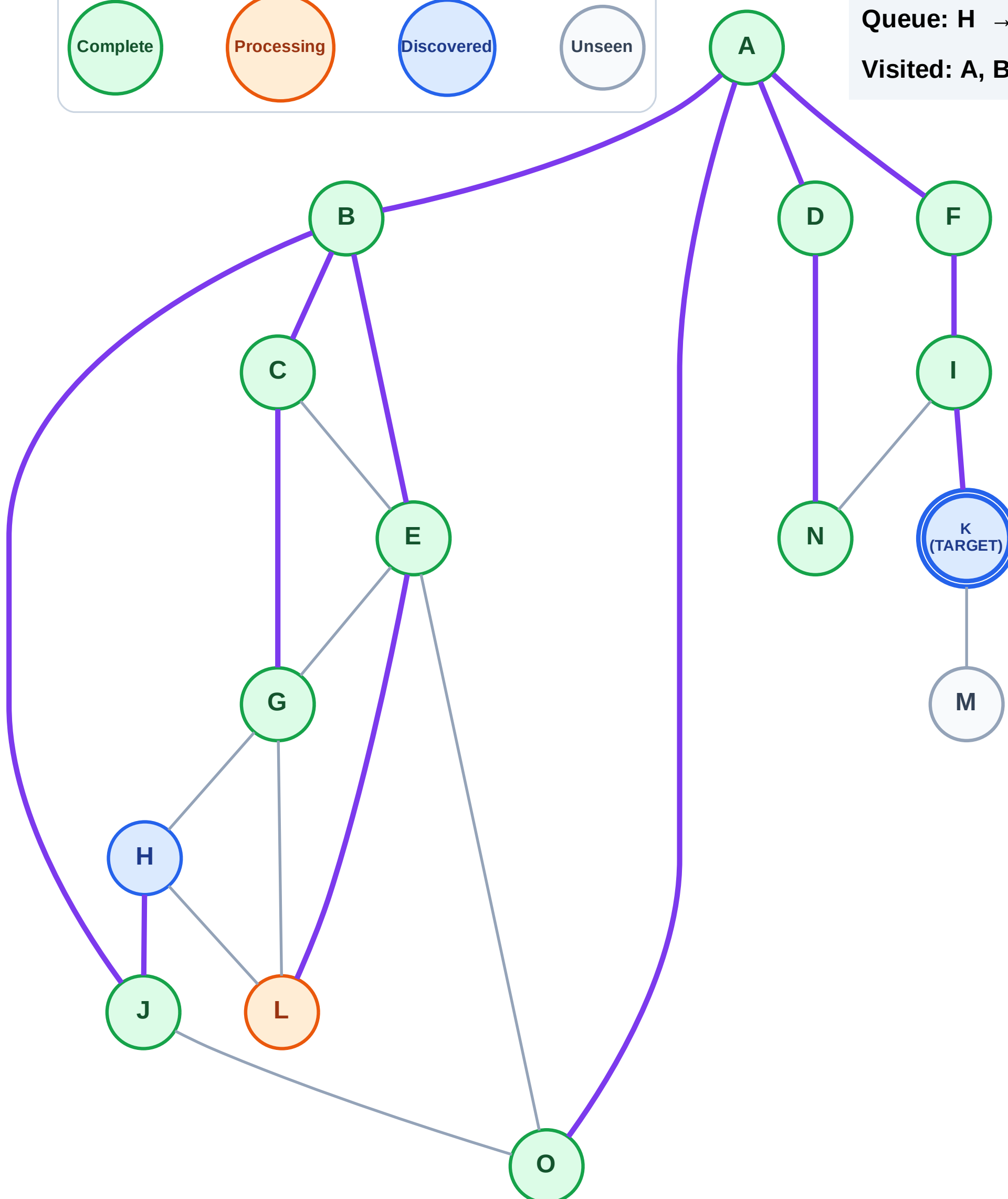
Discovered



Unseen

Queue: H → K

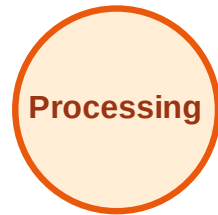
Visited: A, B, C, D, E, F, G, I, J, N, O



# BFS Traversal

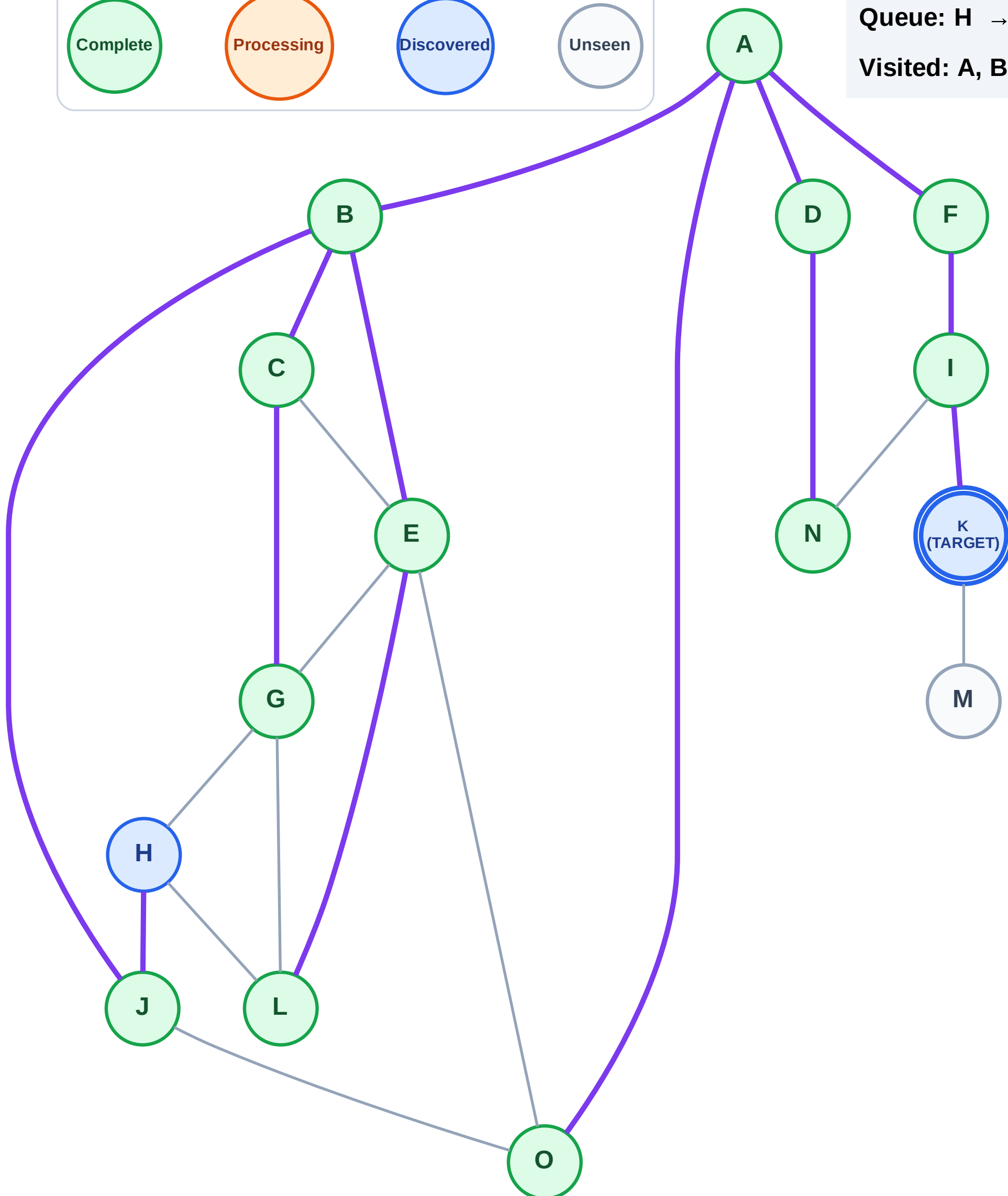
Step 25: No new neighbors from L

## Legend



Queue: H → K

Visited: A, B, C, D, E, F, G, I, J, L, N, O



# BFS Traversal

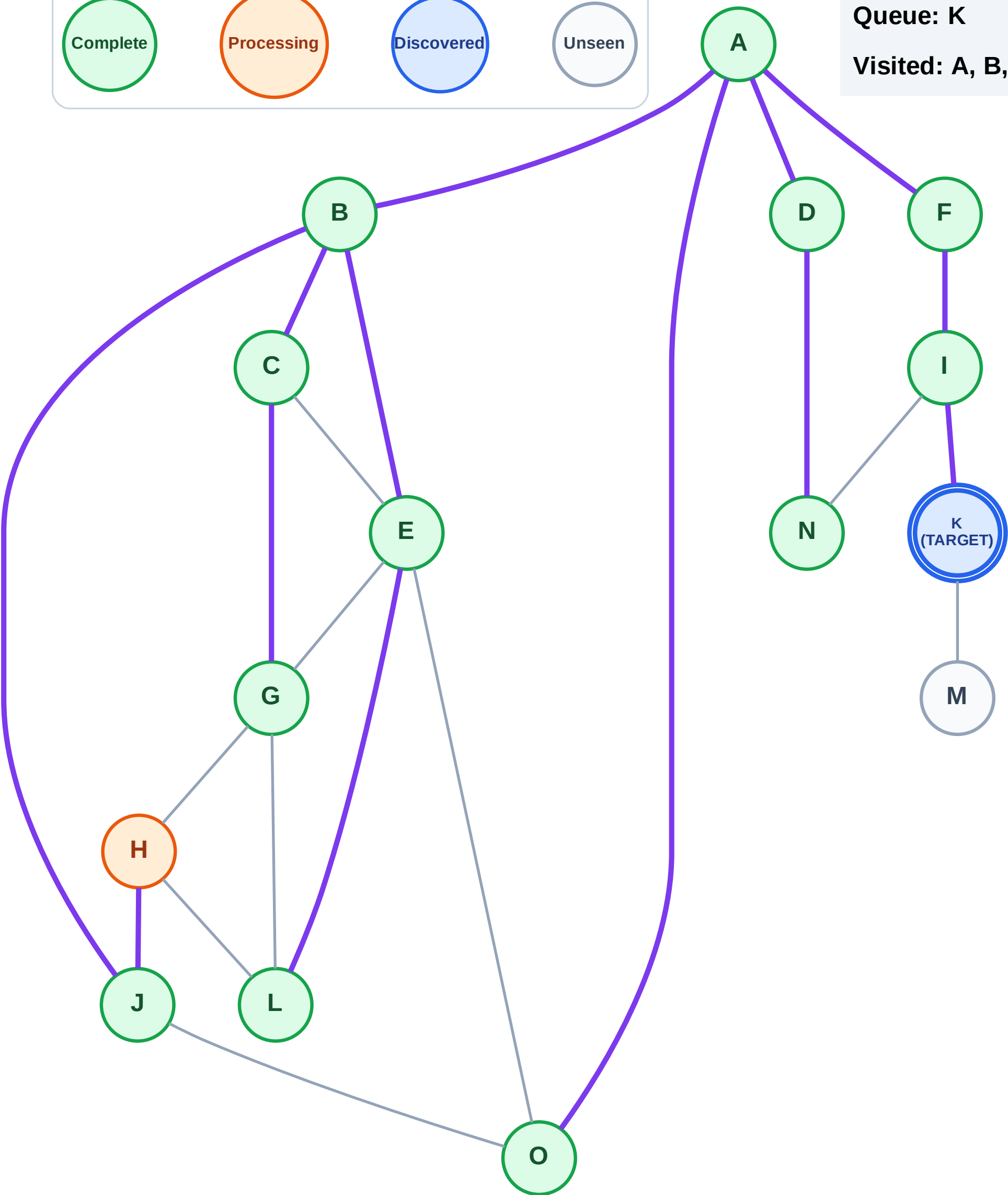
Step 26: Process node H

## Legend



Queue: K

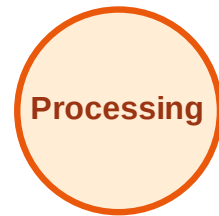
Visited: A, B, C, D, E, F, G, I, J, L, N, O



# BFS Traversal

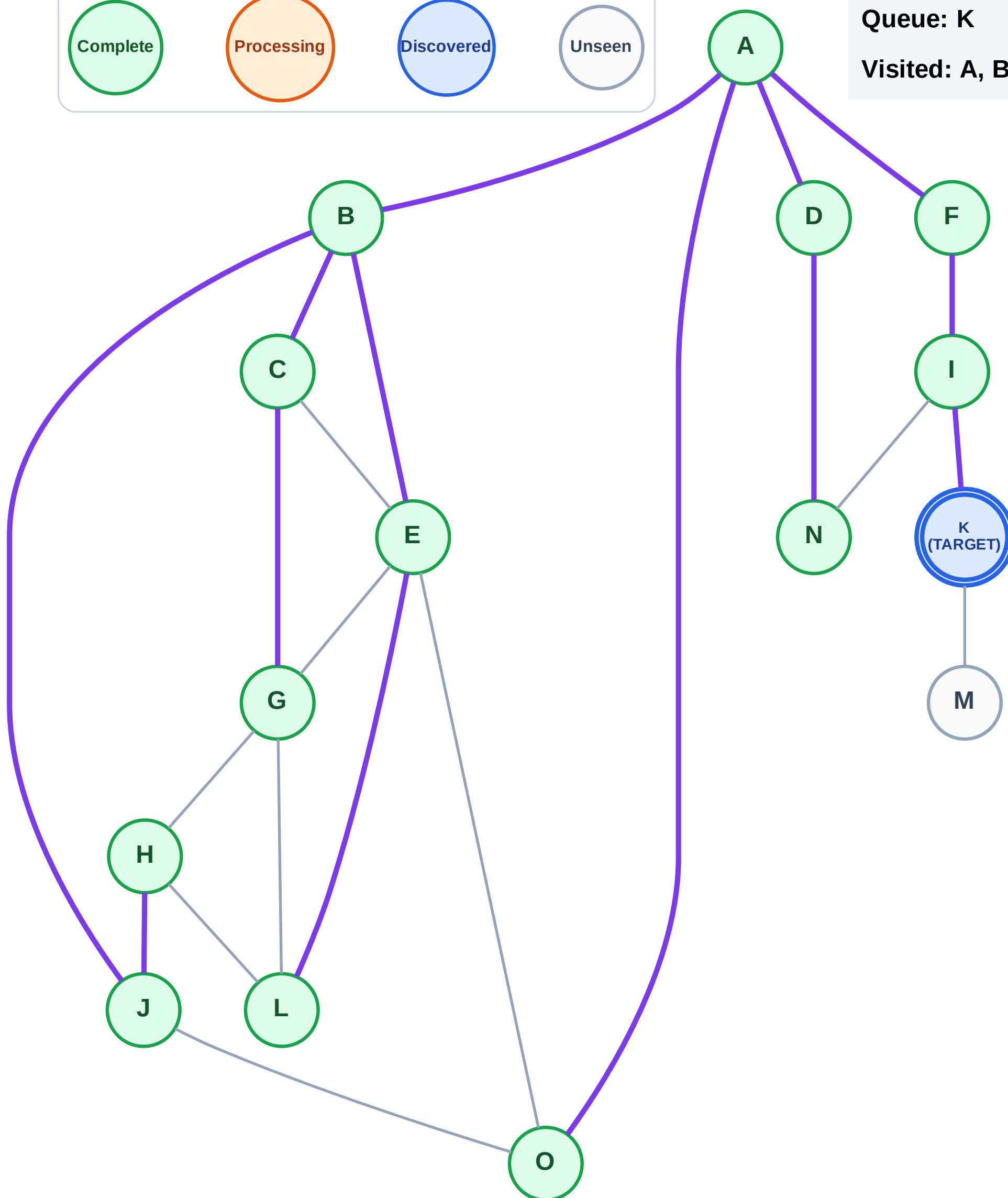
Step 27: No new neighbors from H

## Legend



Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, N, O



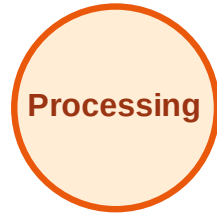
# BFS Traversal

Step 28: Process node K

## Legend



Complete



Processing



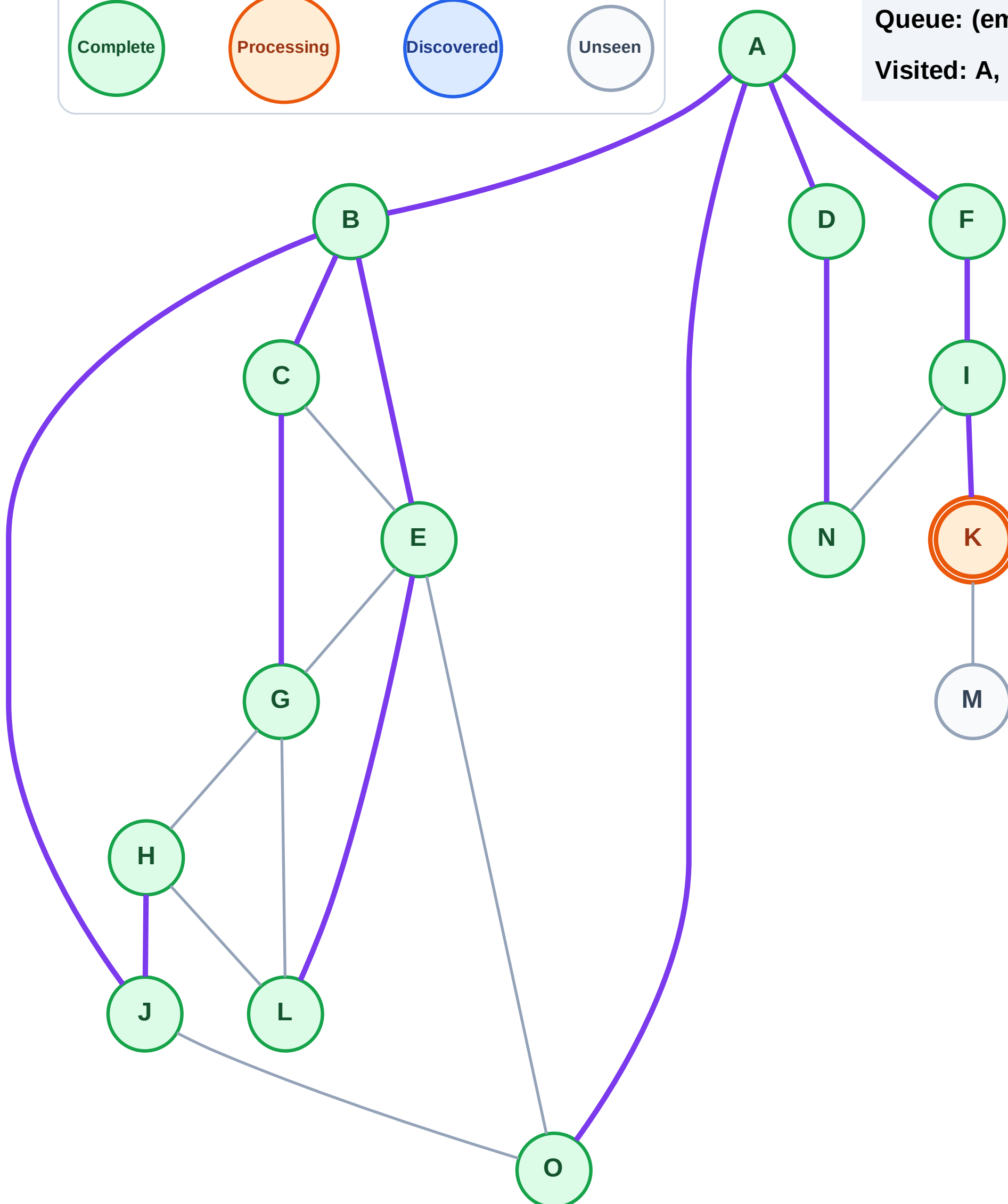
Discovered



Unseen

Queue: (empty)

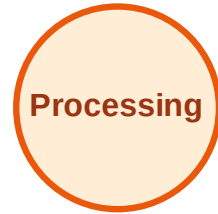
Visited: A, B, C, D, E, F, G, H, I, J, L, N, O



# BFS Traversal

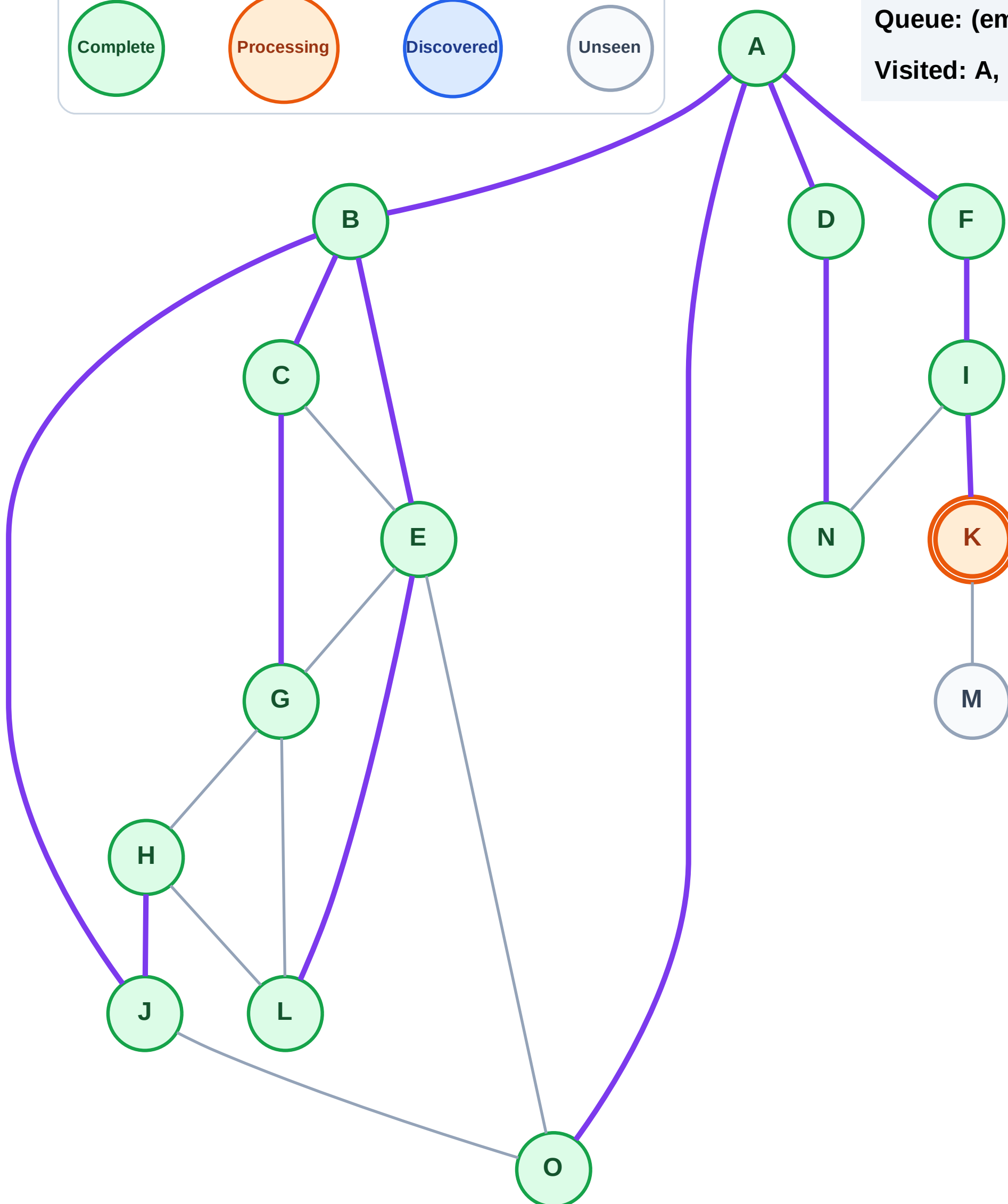
Step 29: TARGET NODE K REACHED! Path (3 hops): A → F → I → K

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, N, O



# DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

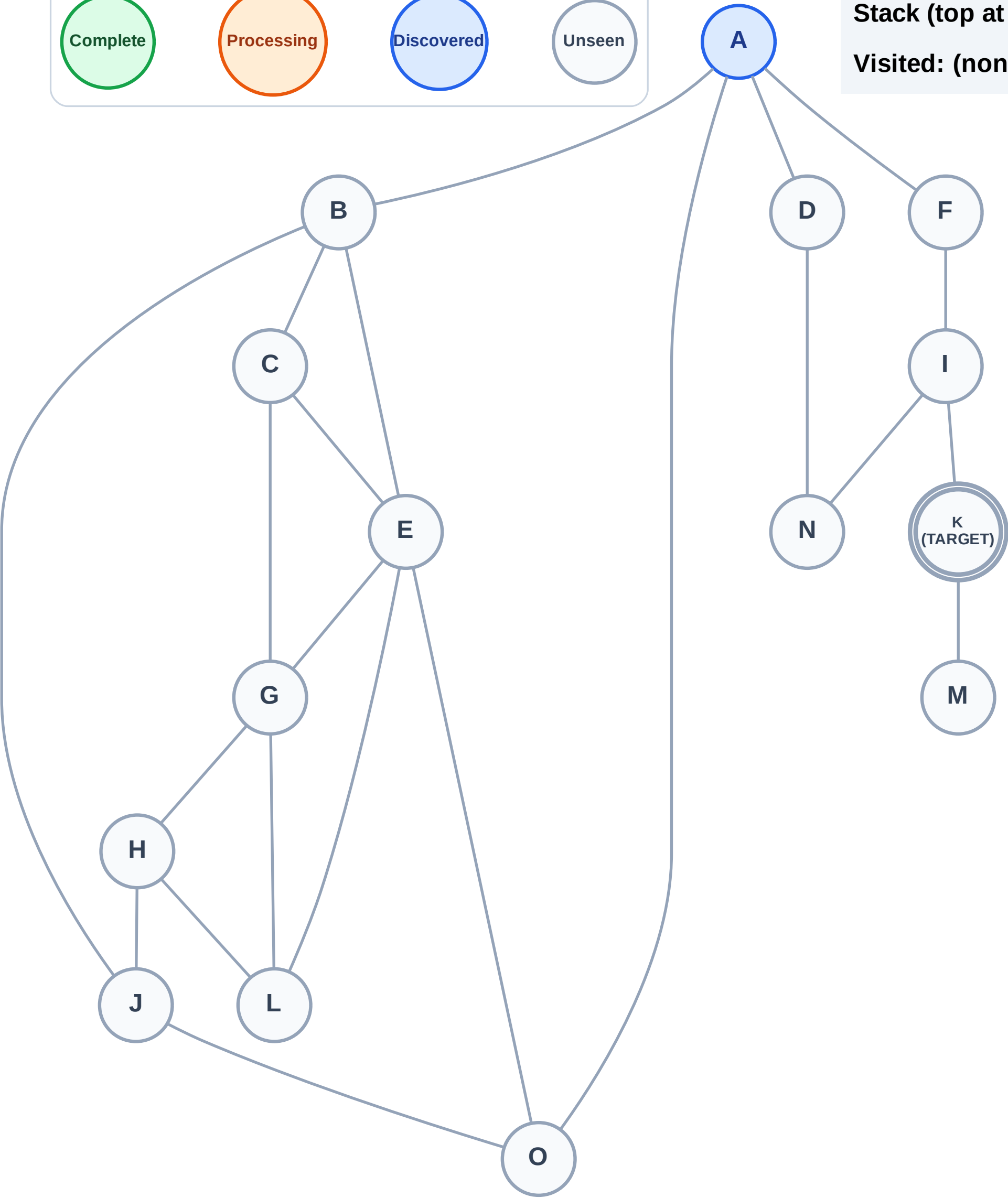
Complete

Processing

Discovered

Unseen

Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

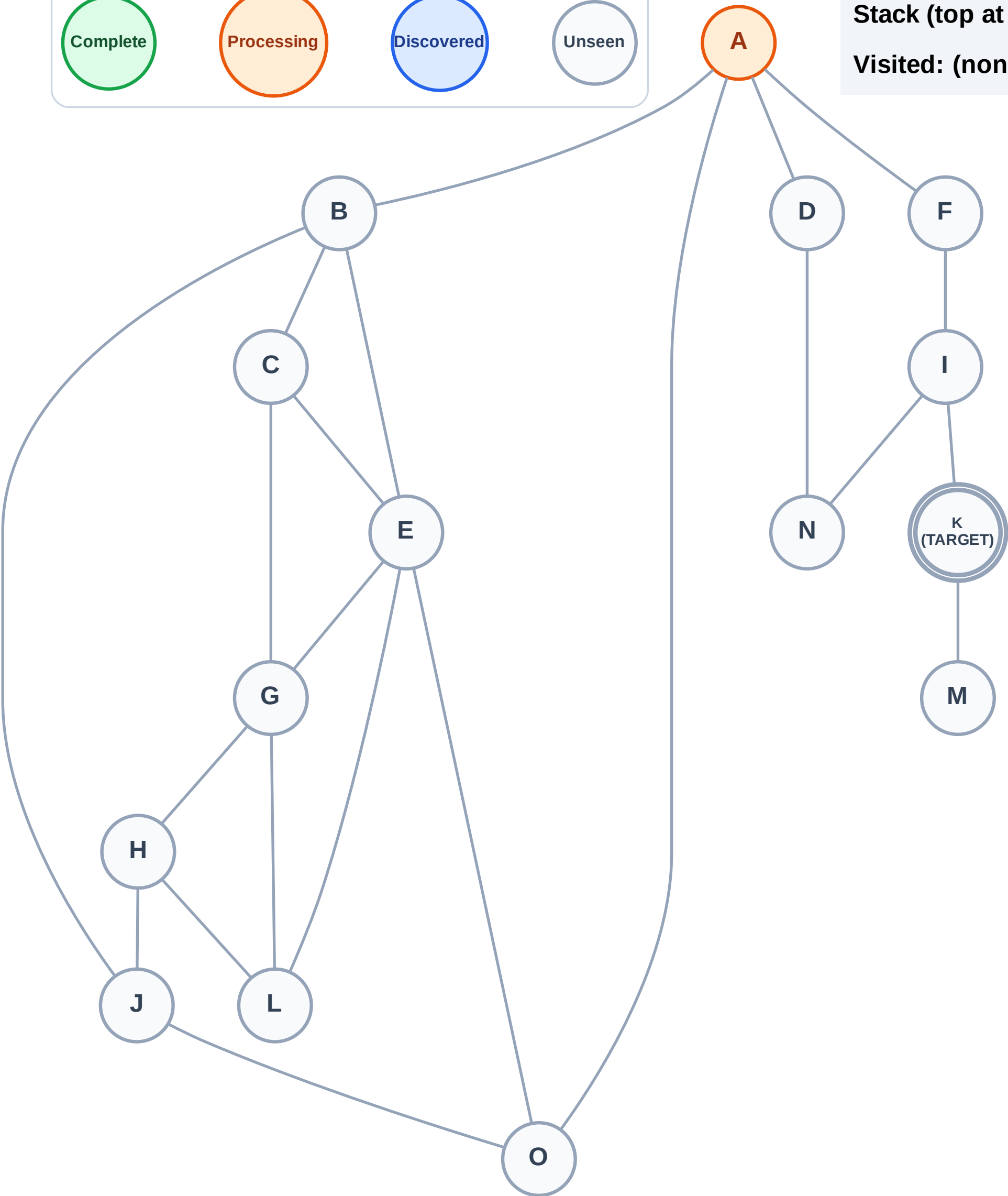
Complete

Processing

Discovered

Unseen

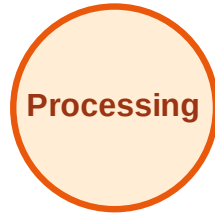
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

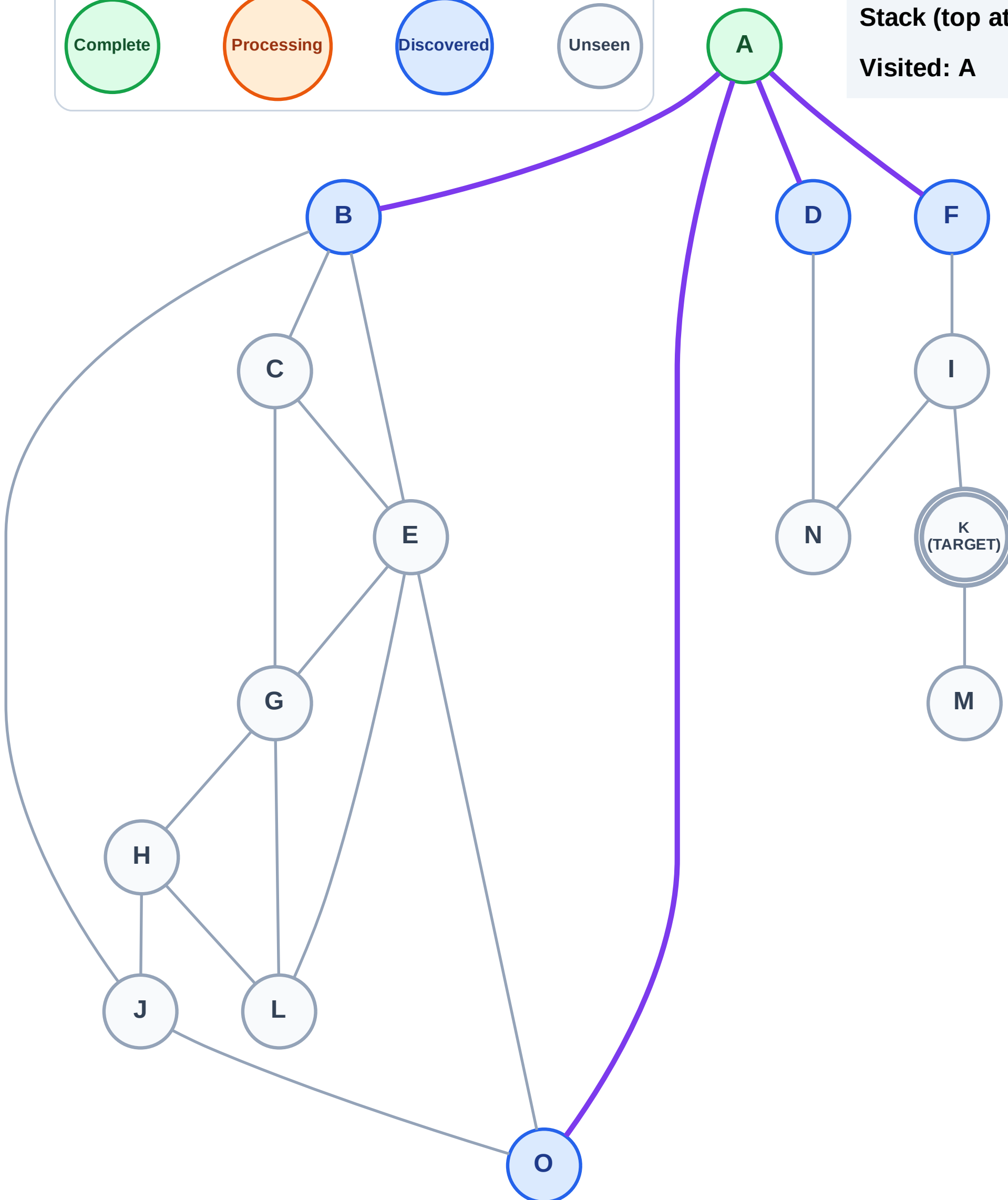
Step 3: Discover O, F, D, B from A

## Legend



Stack (top at right): O | F | D | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

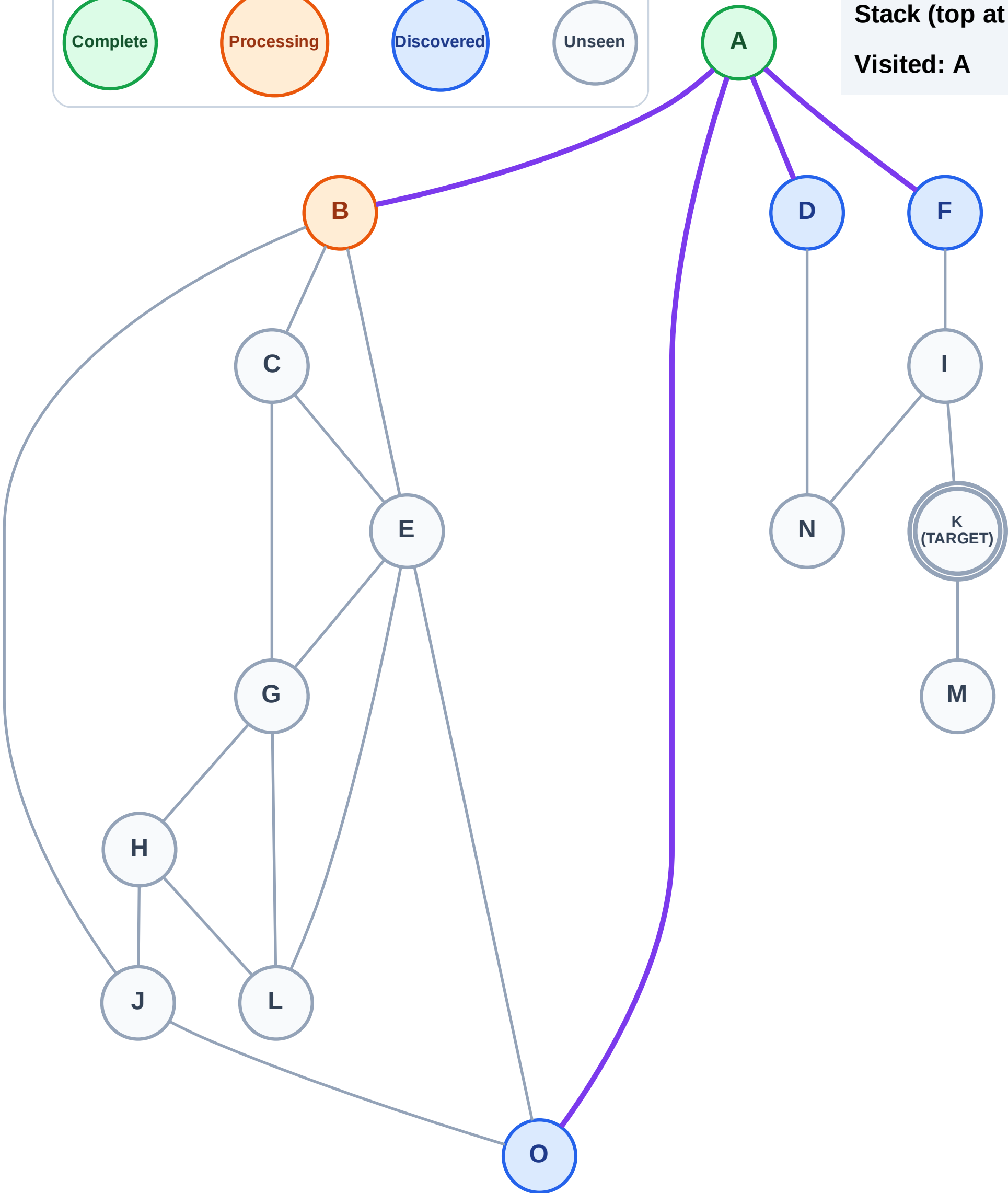
Processing

Discovered

Unseen

Stack (top at right): O | F | D

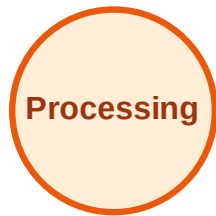
Visited: A



# DFS Traversal

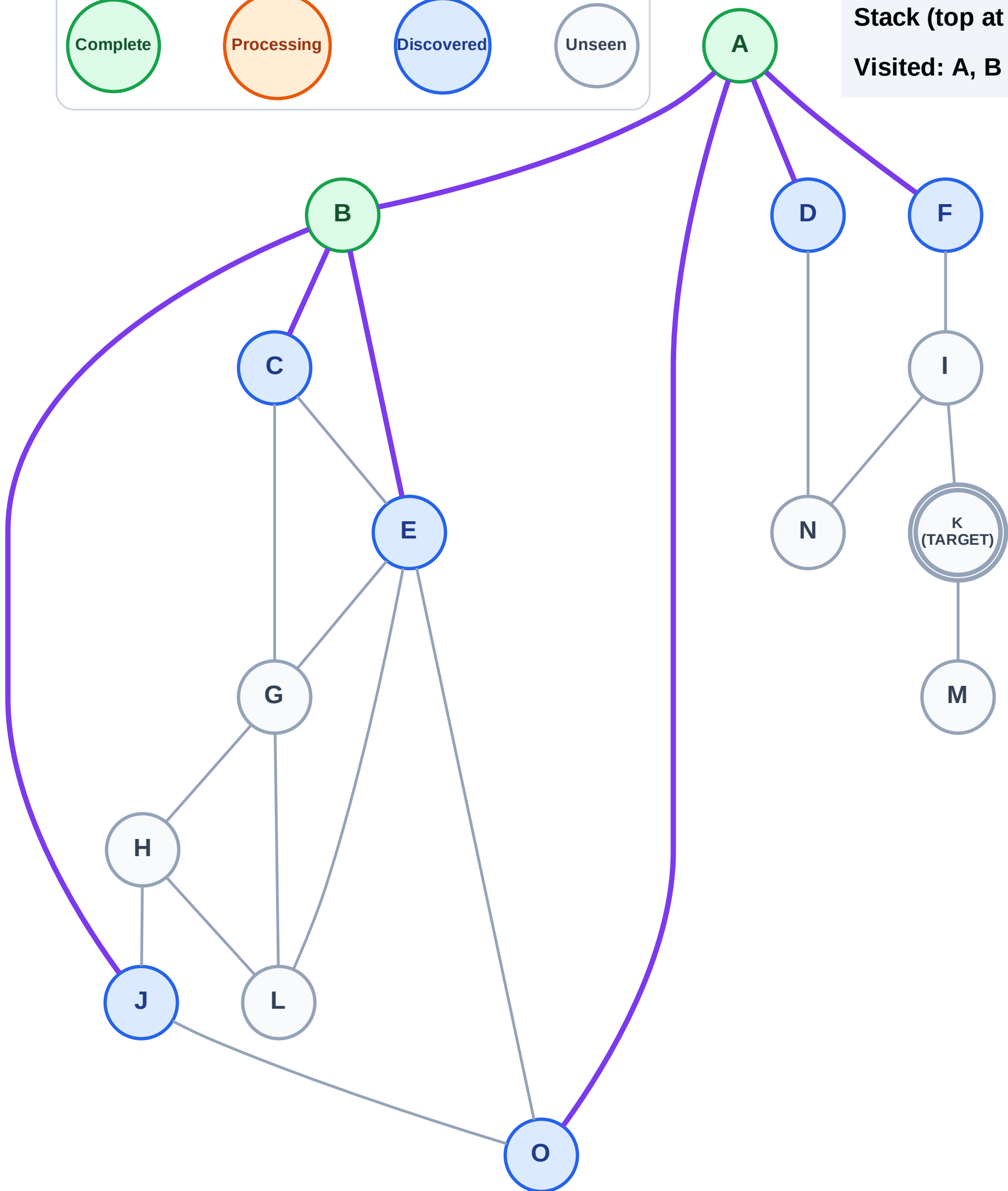
Step 5: Discover J, E, C from B

## Legend



Stack (top at right): O | F | D | J | E | C

Visited: A, B



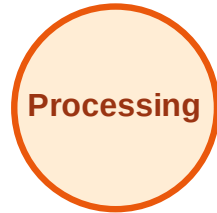
# DFS Traversal

Step 6: Process node C

## Legend



Complete



Processing



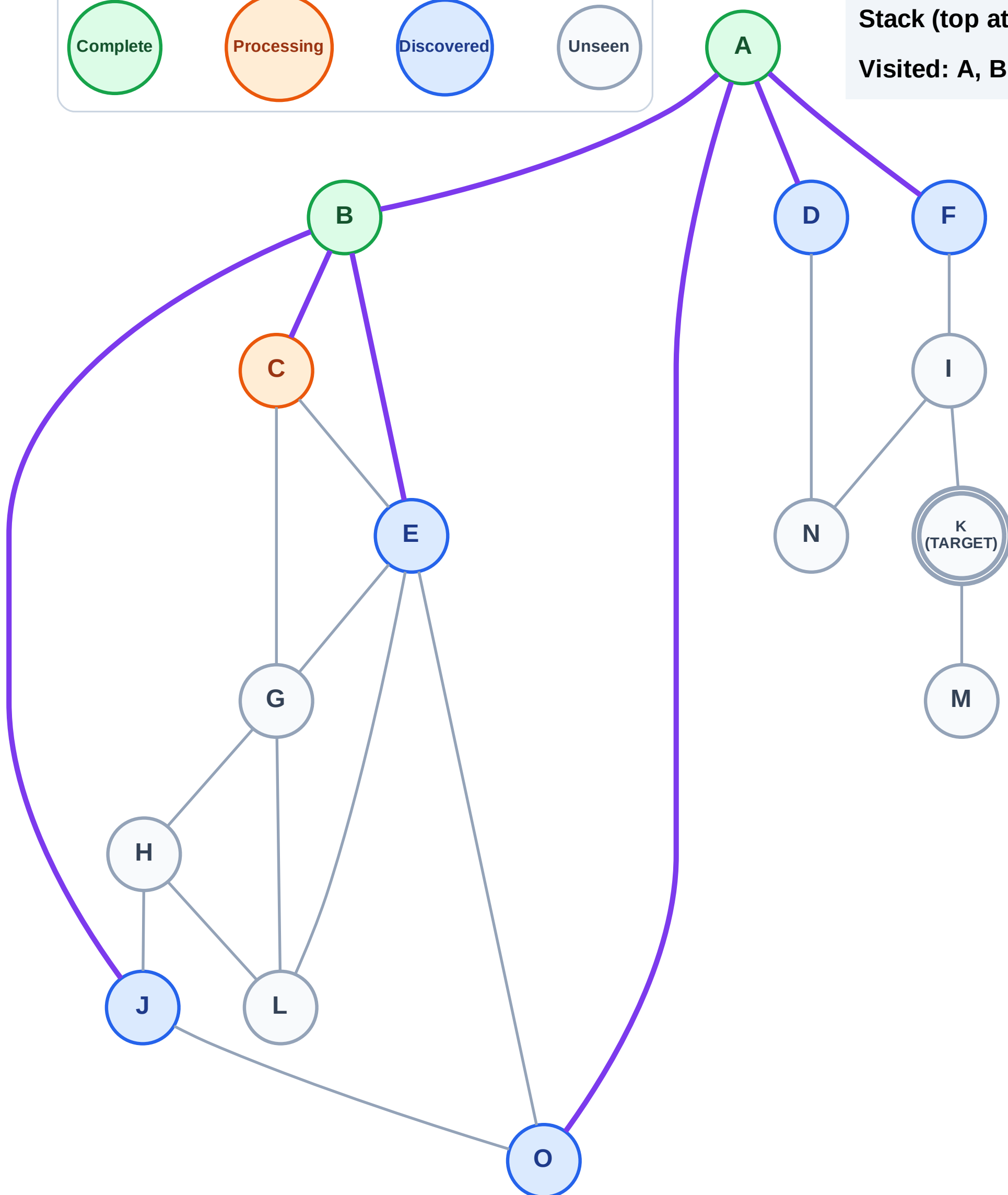
Discovered



Unseen

Stack (top at right): O | F | D | J | E

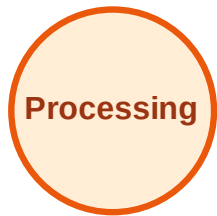
Visited: A, B



# DFS Traversal

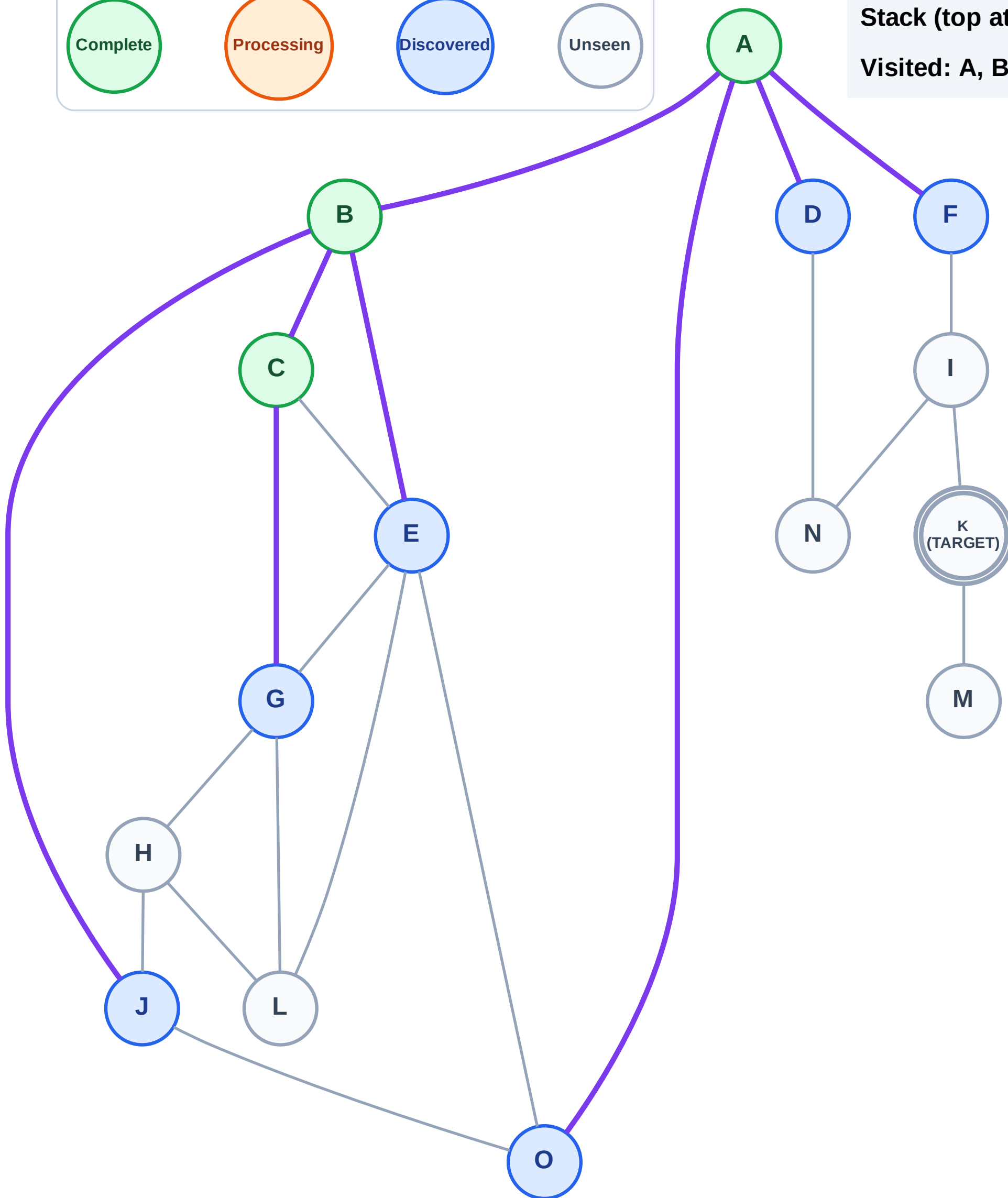
Step 7: Discover G from C

## Legend



Stack (top at right): O | F | D | J | E | G

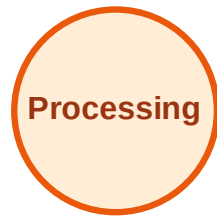
Visited: A, B, C



# DFS Traversal

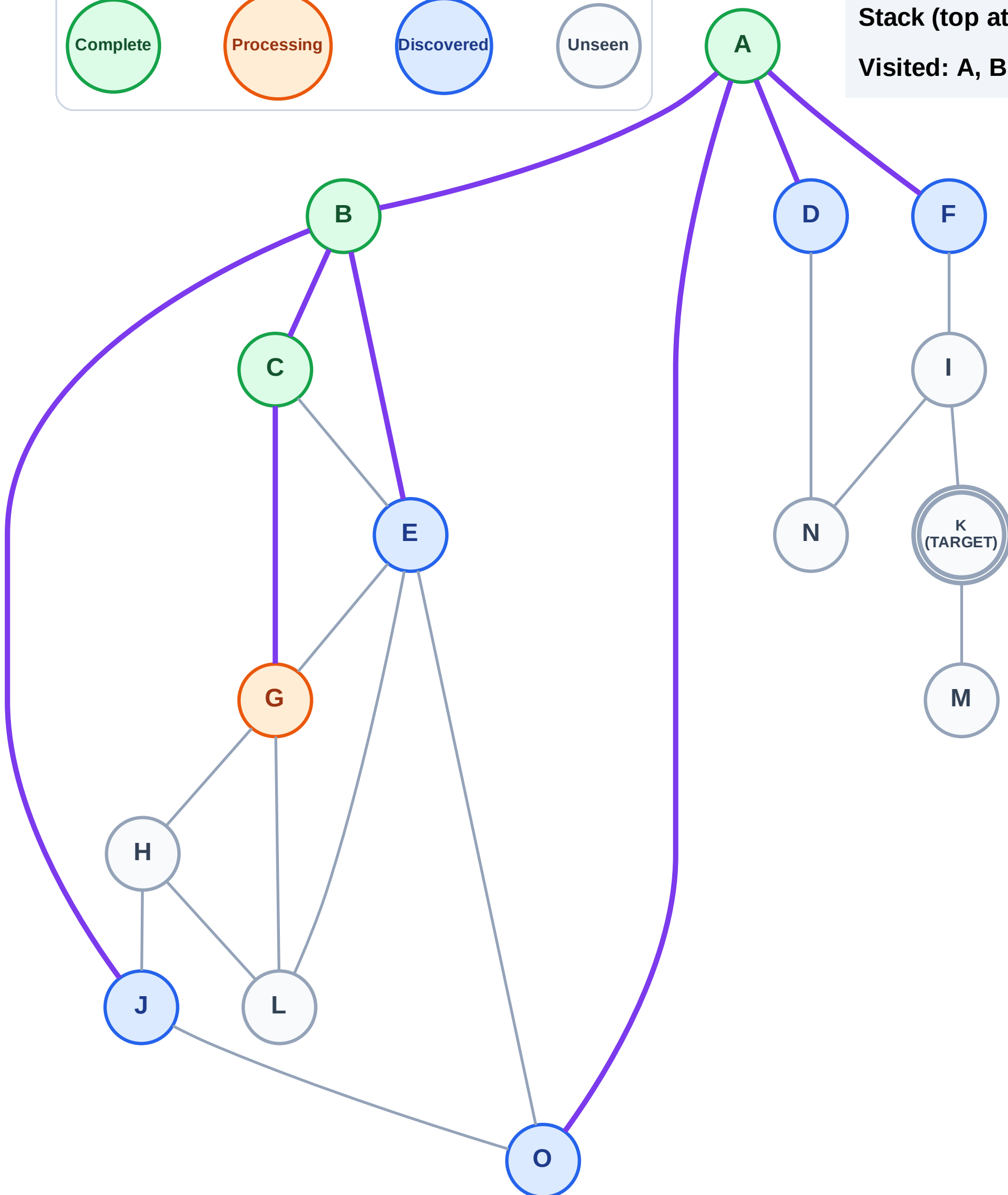
Step 8: Process node G

## Legend



Stack (top at right): O | F | D | J | E

Visited: A, B, C



# DFS Traversal

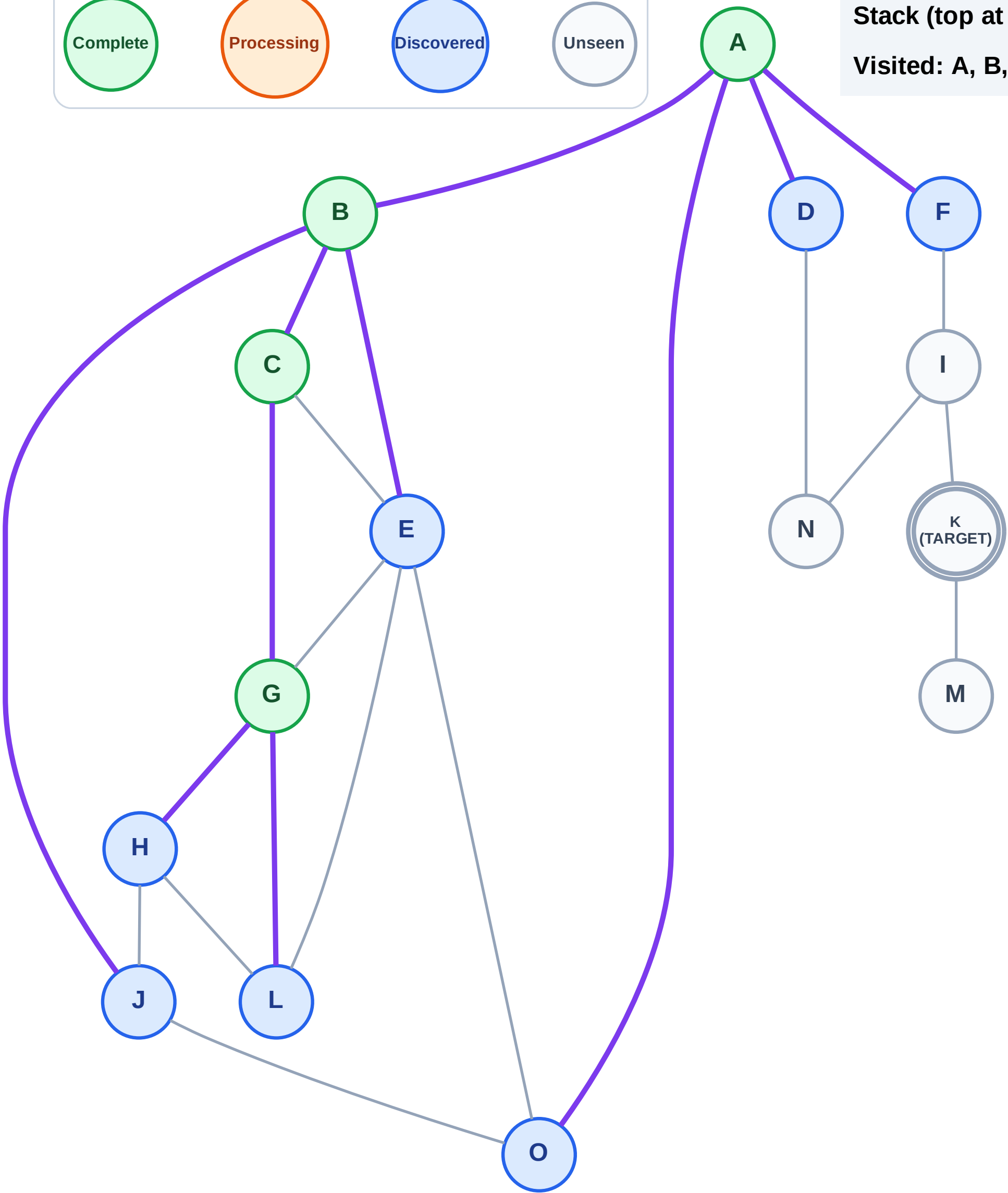
Step 9: Discover L, H from G

## Legend



Stack (top at right): O | F | D | J | E | L | H

Visited: A, B, C, G



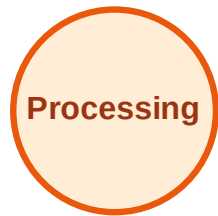
# DFS Traversal

Step 10: Process node H

## Legend



Complete



Processing



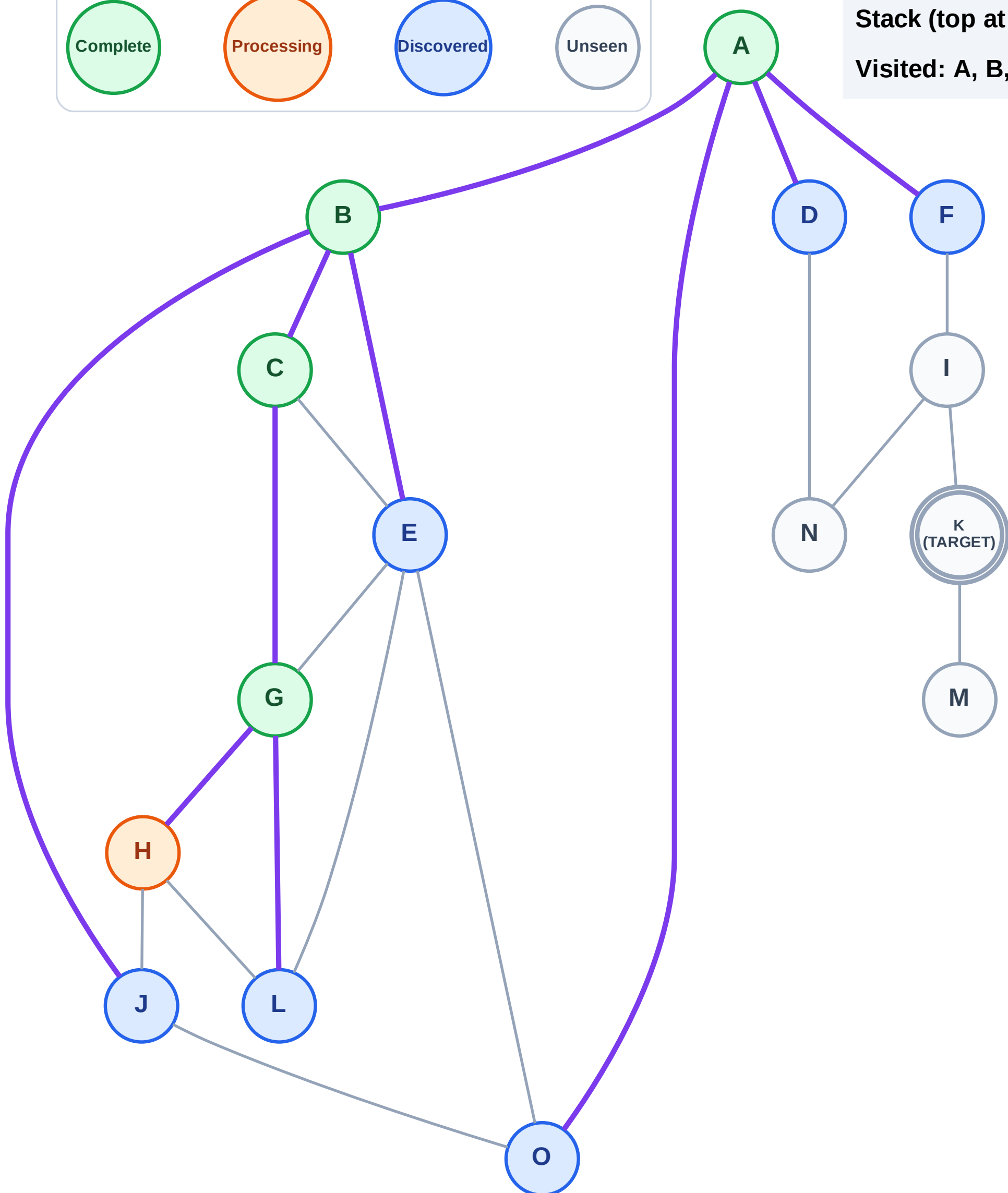
Discovered



Unseen

Stack (top at right): O | F | D | J | E | L

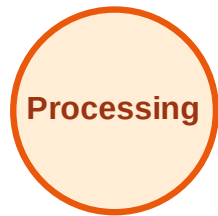
Visited: A, B, C, G



# DFS Traversal

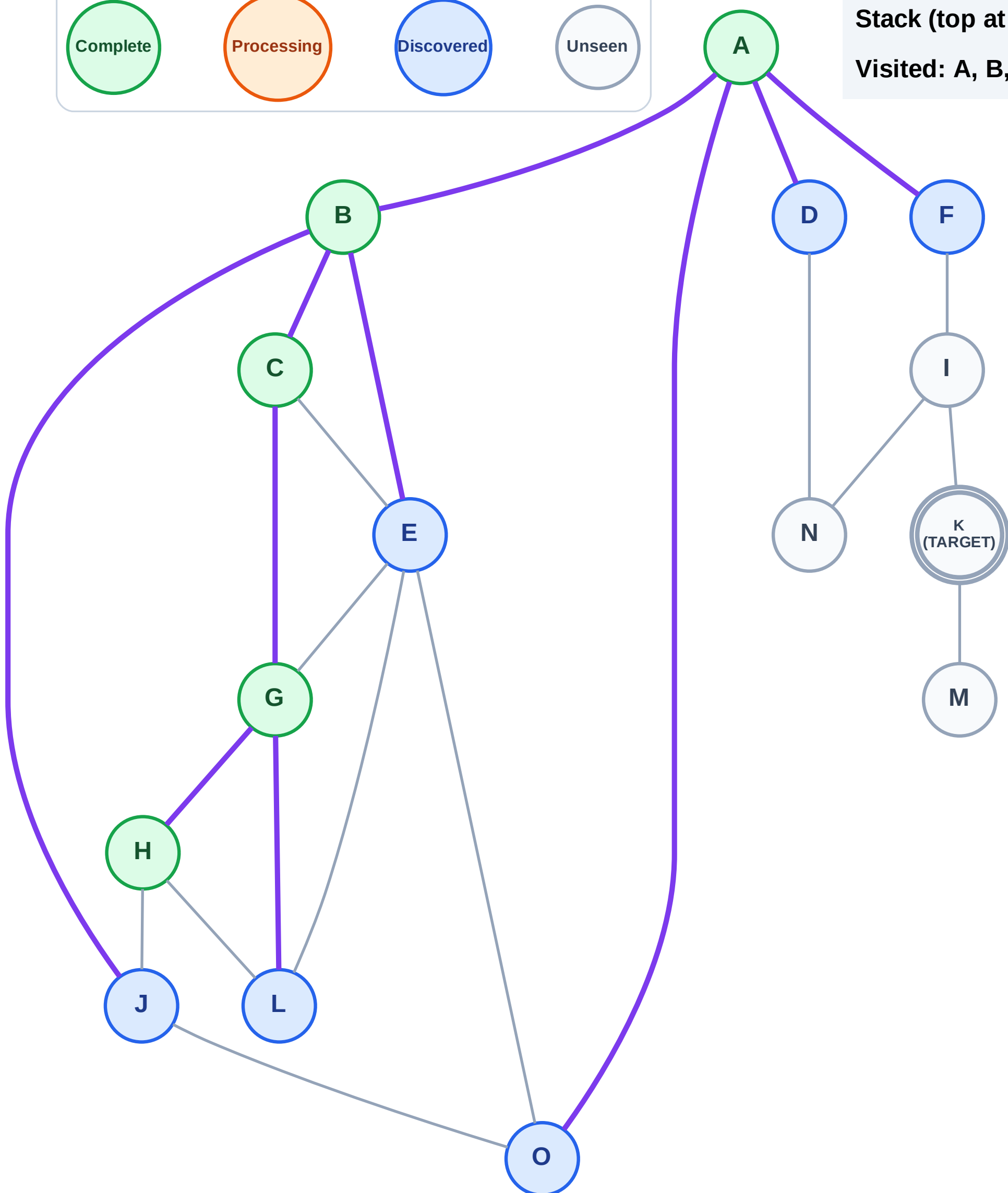
Step 11: No new neighbors from H

## Legend



Stack (top at right): O | F | D | J | E | L

Visited: A, B, C, G, H

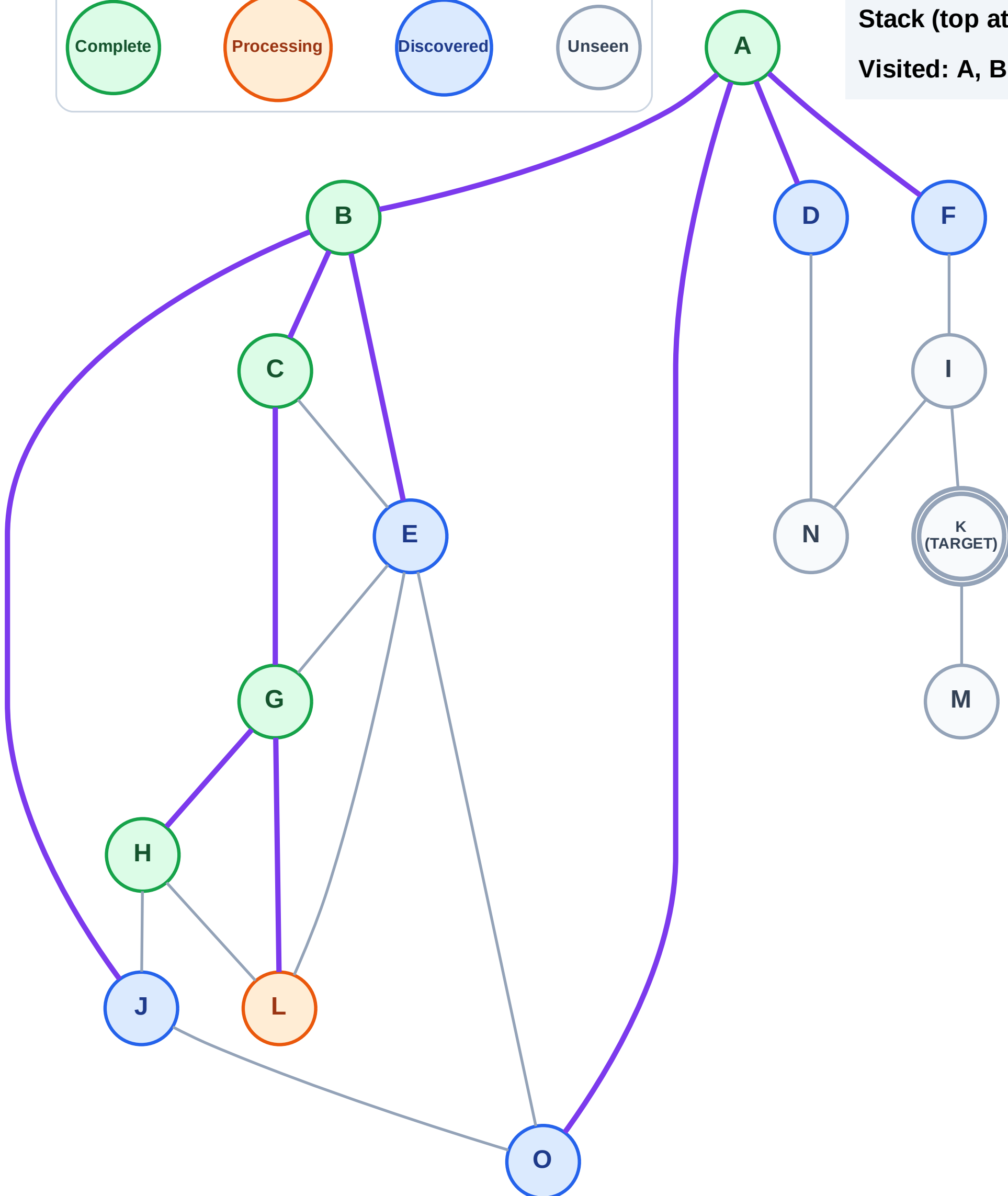


### Step 12: Process node L

### Legend

## Unseen

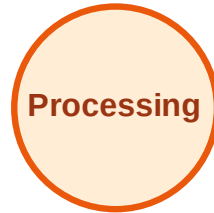
**Visited: A, B, C, G, H**



# DFS Traversal

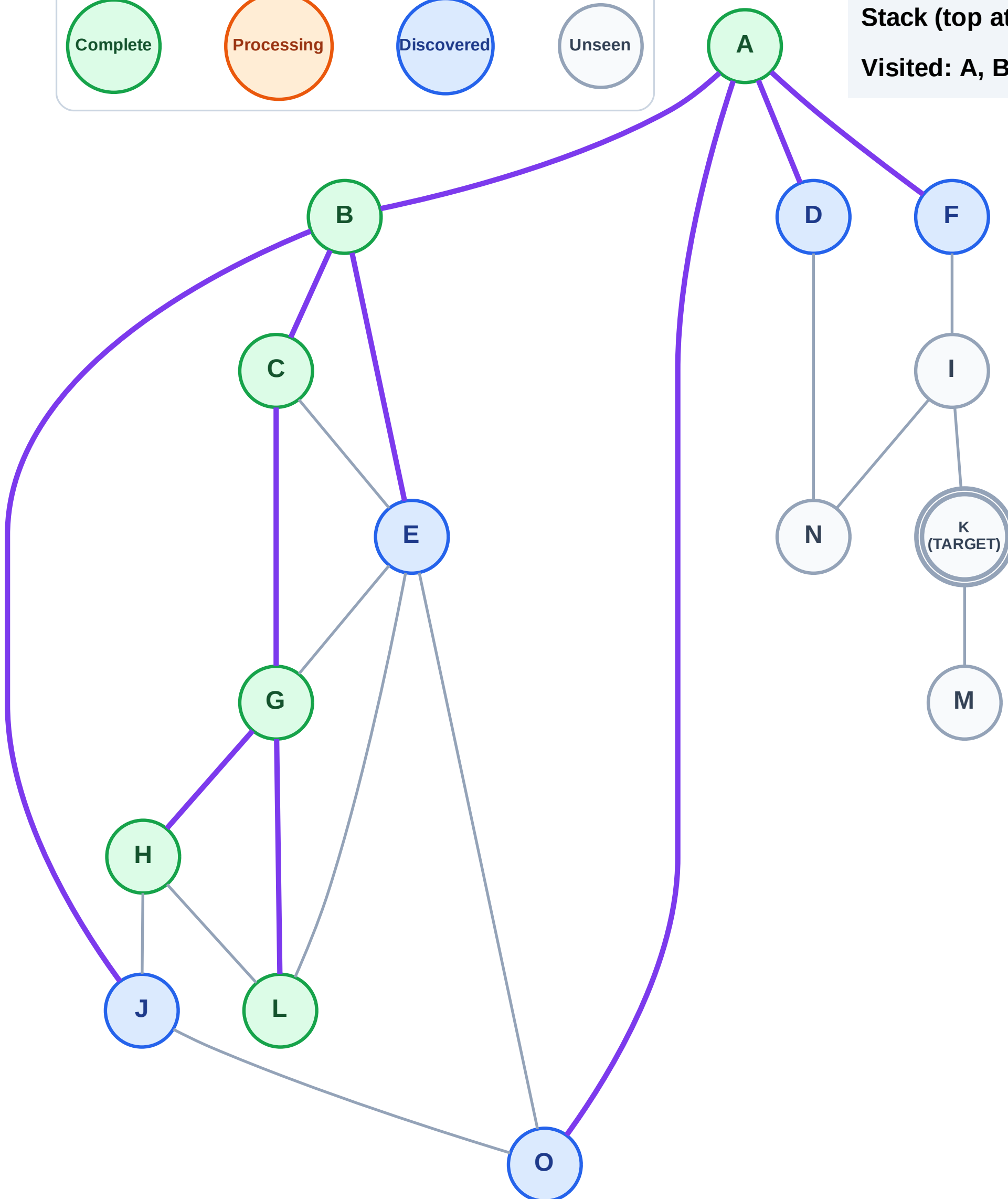
Step 13: No new neighbors from L

## Legend



Stack (top at right): O | F | D | J | E

Visited: A, B, C, G, H, L



# DFS Traversal

Step 14: Process node E

## Legend



Complete



Processing



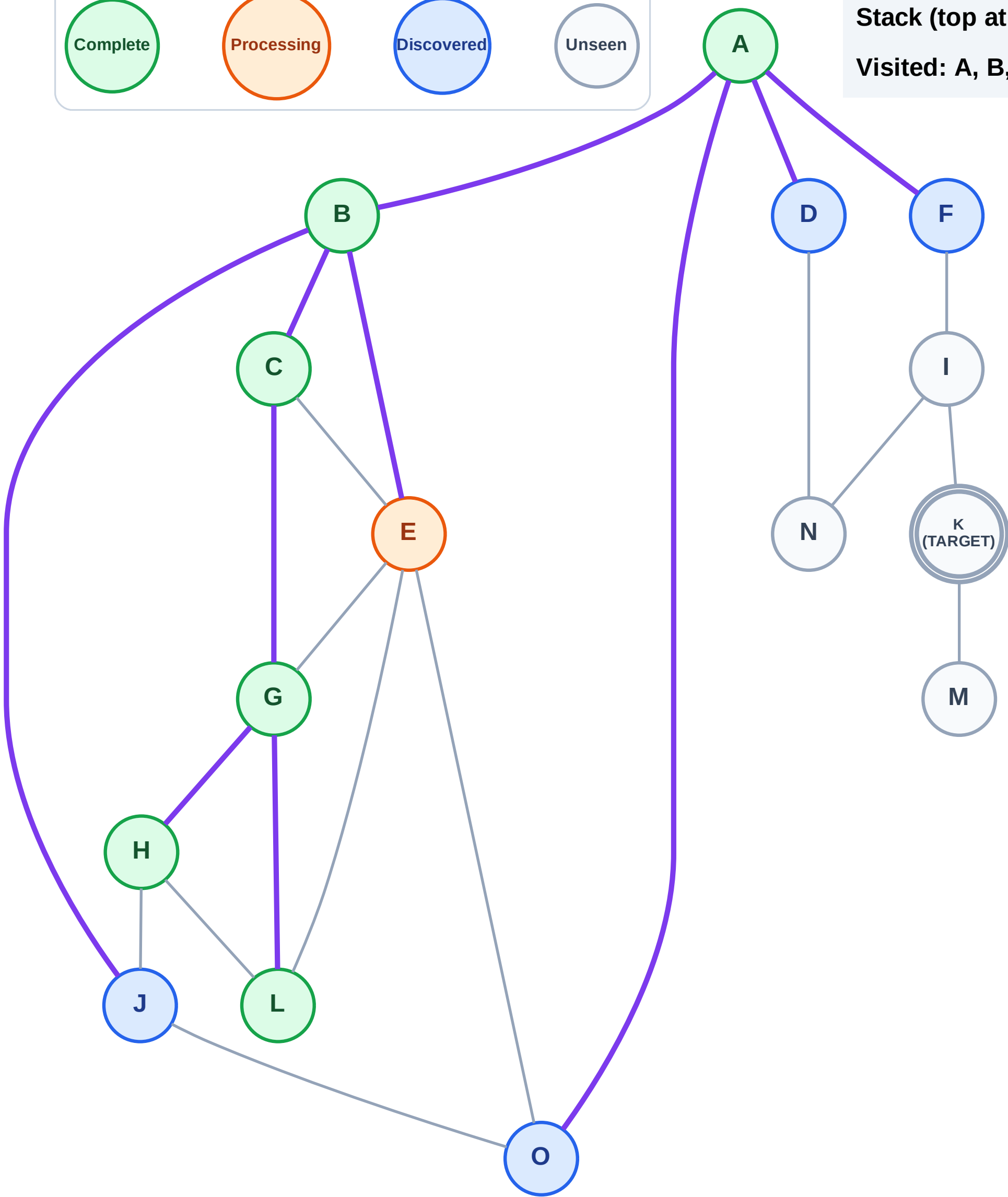
Discovered



Unseen

Stack (top at right): O | F | D | J

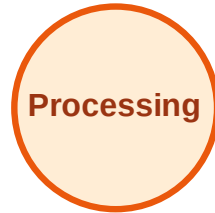
Visited: A, B, C, G, H, L



# DFS Traversal

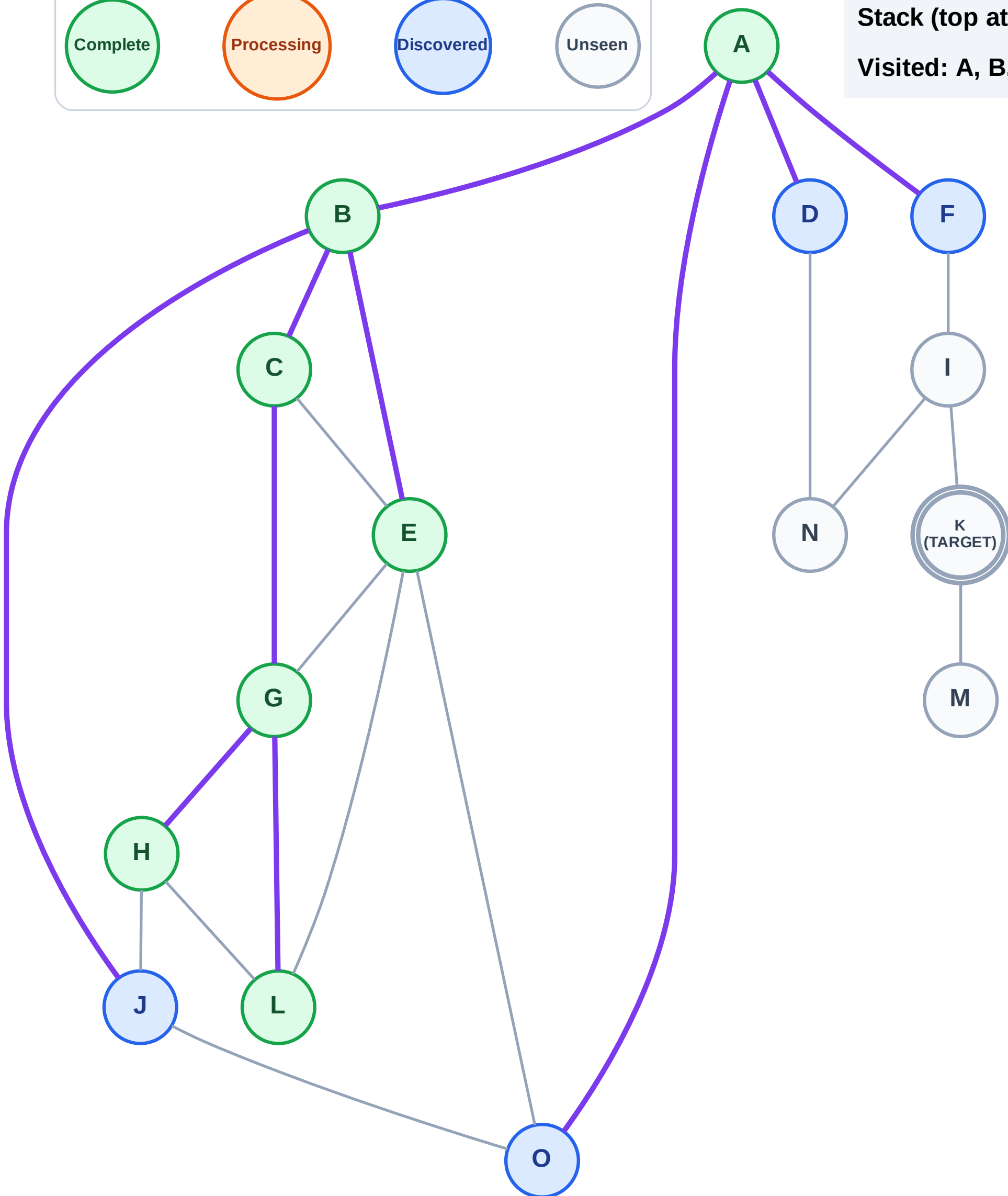
Step 15: No new neighbors from E

## Legend



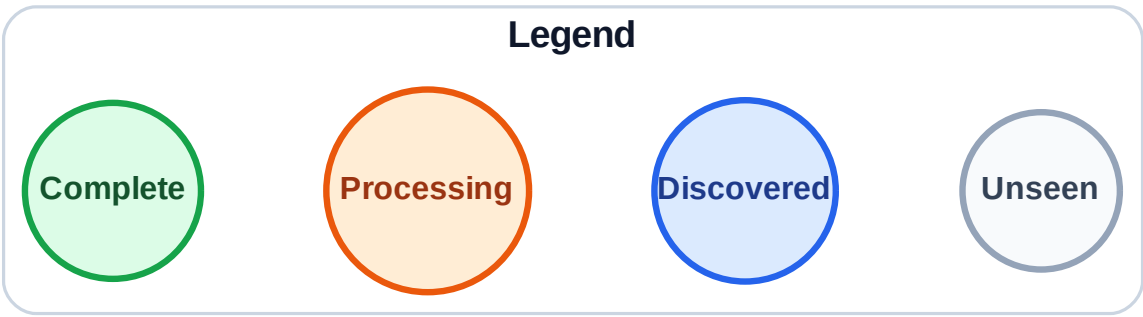
Stack (top at right): O | F | D | J

Visited: A, B, C, E, G, H, L



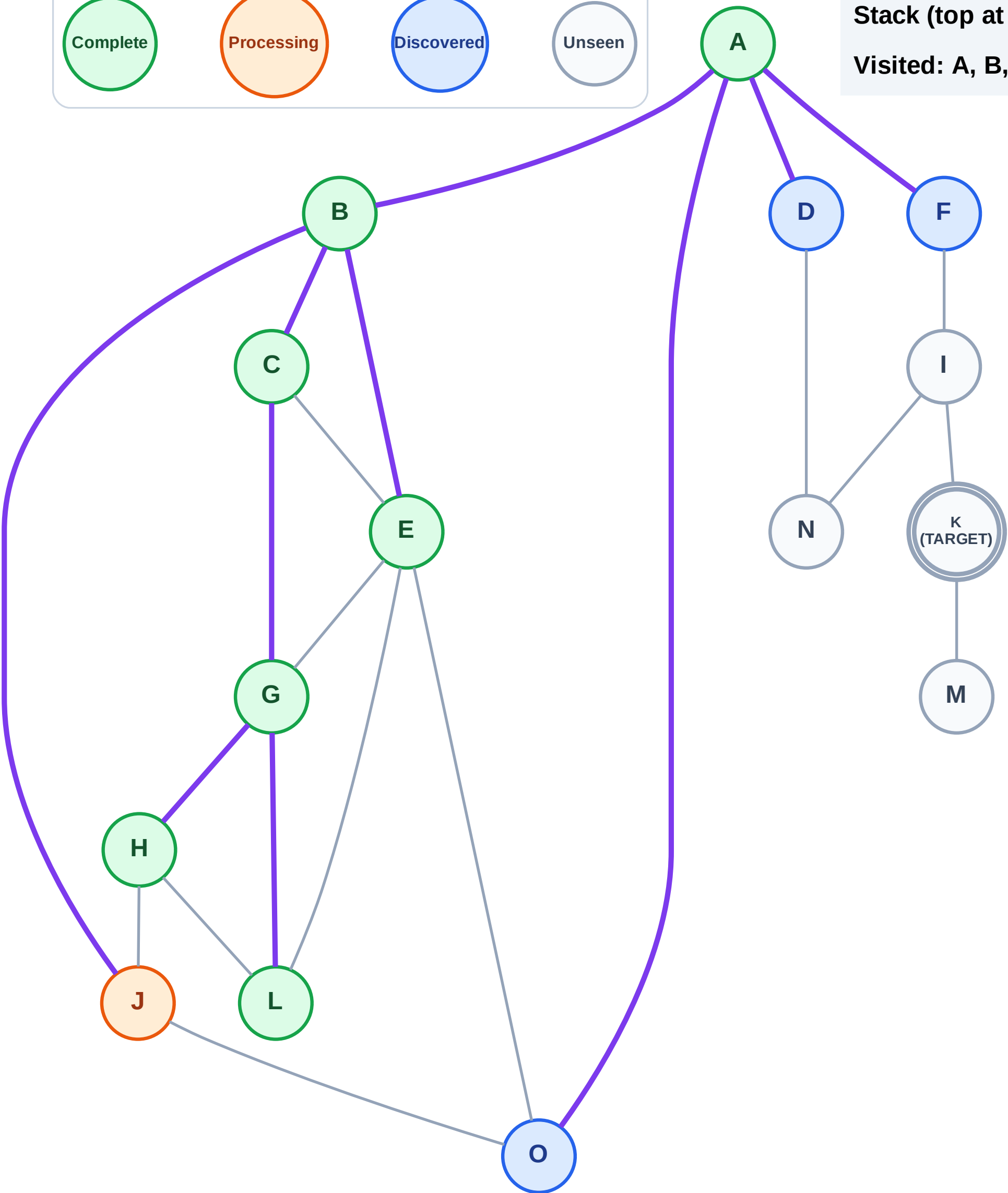
# DFS Traversal

Step 16: Process node J



Stack (top at right): O | F | D

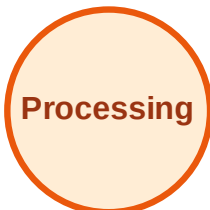
Visited: A, B, C, E, G, H, L



# DFS Traversal

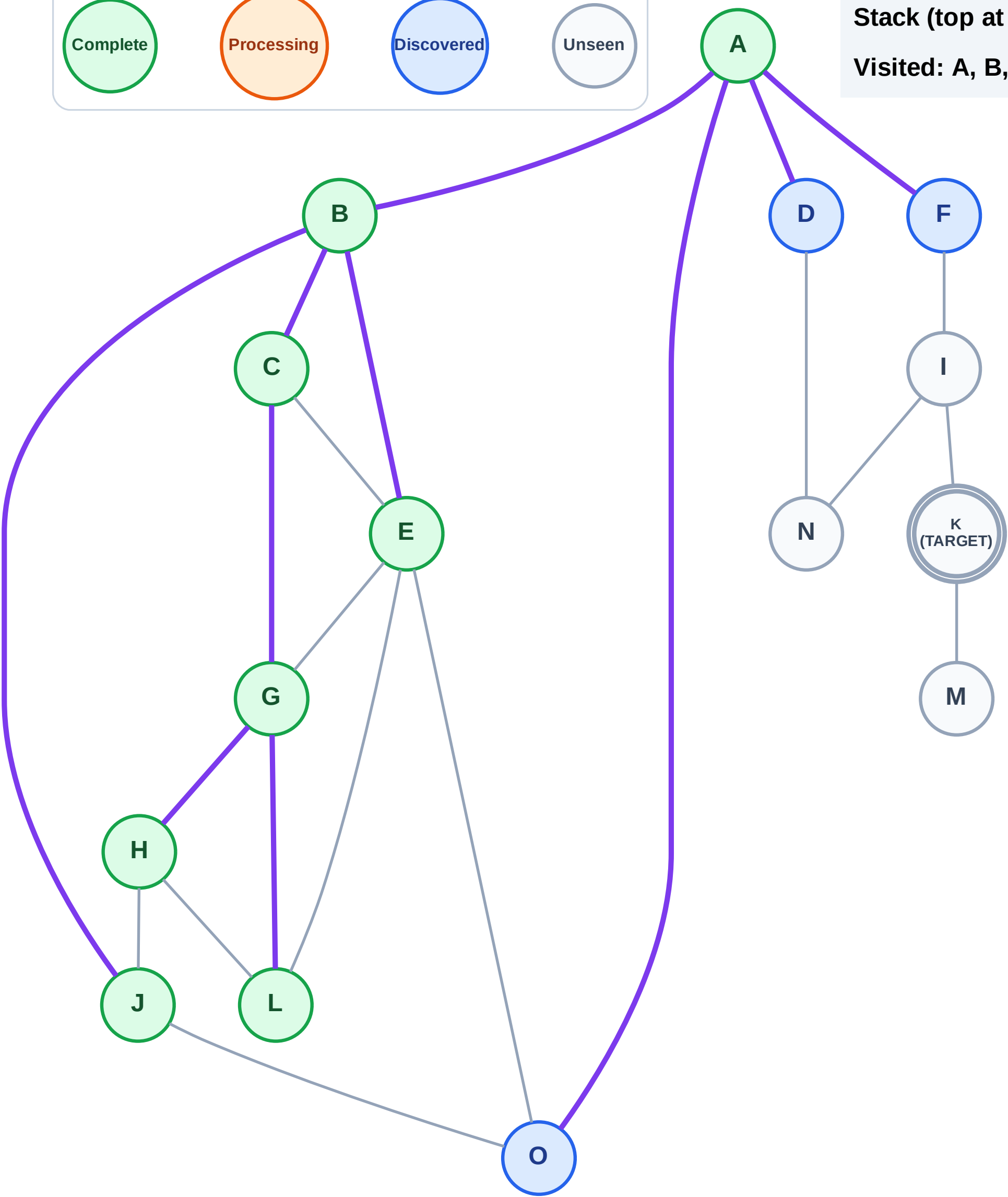
Step 17: No new neighbors from J

## Legend



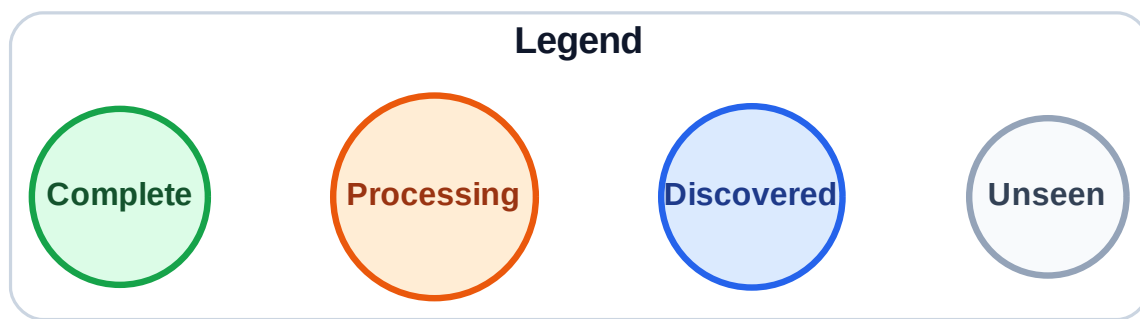
Stack (top at right): O | F | D

Visited: A, B, C, E, G, H, J, L



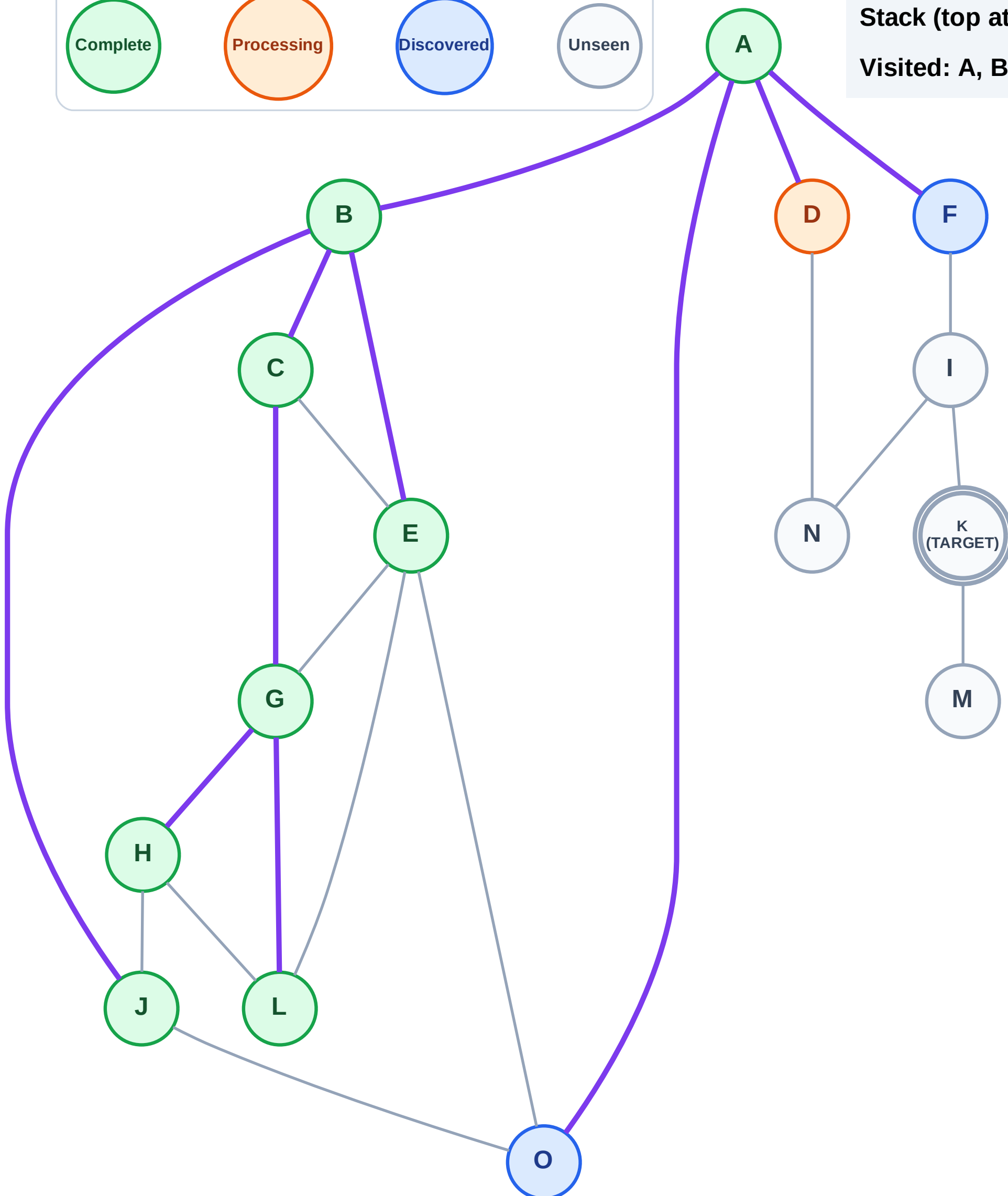
# DFS Traversal

Step 18: Process node D



Stack (top at right): O | F

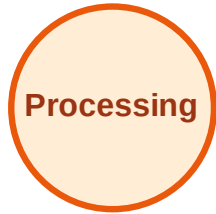
Visited: A, B, C, E, G, H, J, L



# DFS Traversal

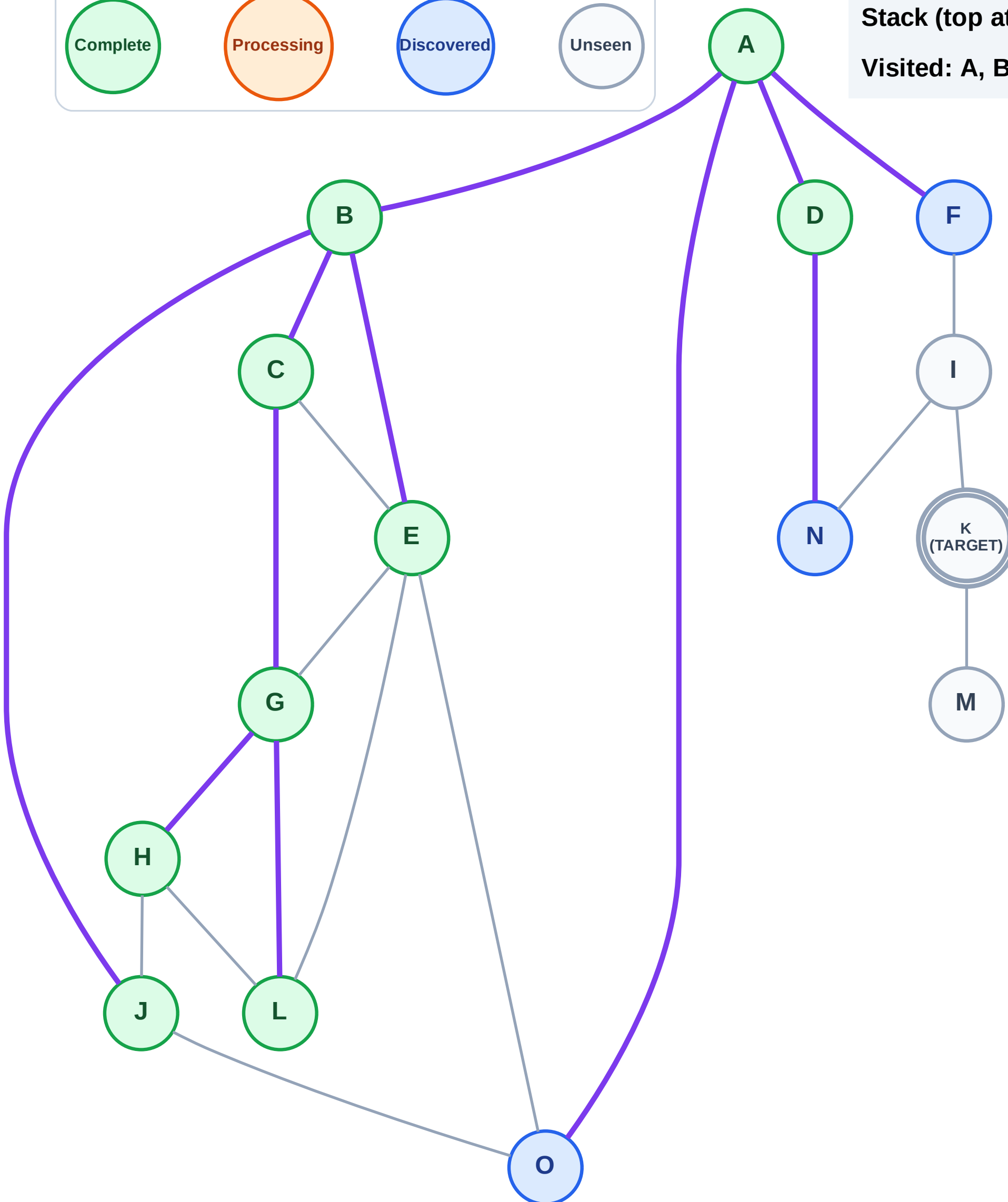
Step 19: Discover N from D

## Legend



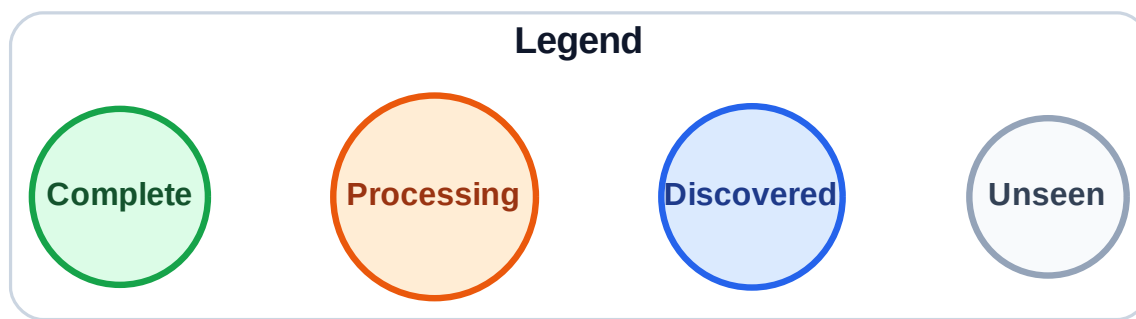
Stack (top at right): O | F | N

Visited: A, B, C, D, E, G, H, J, L

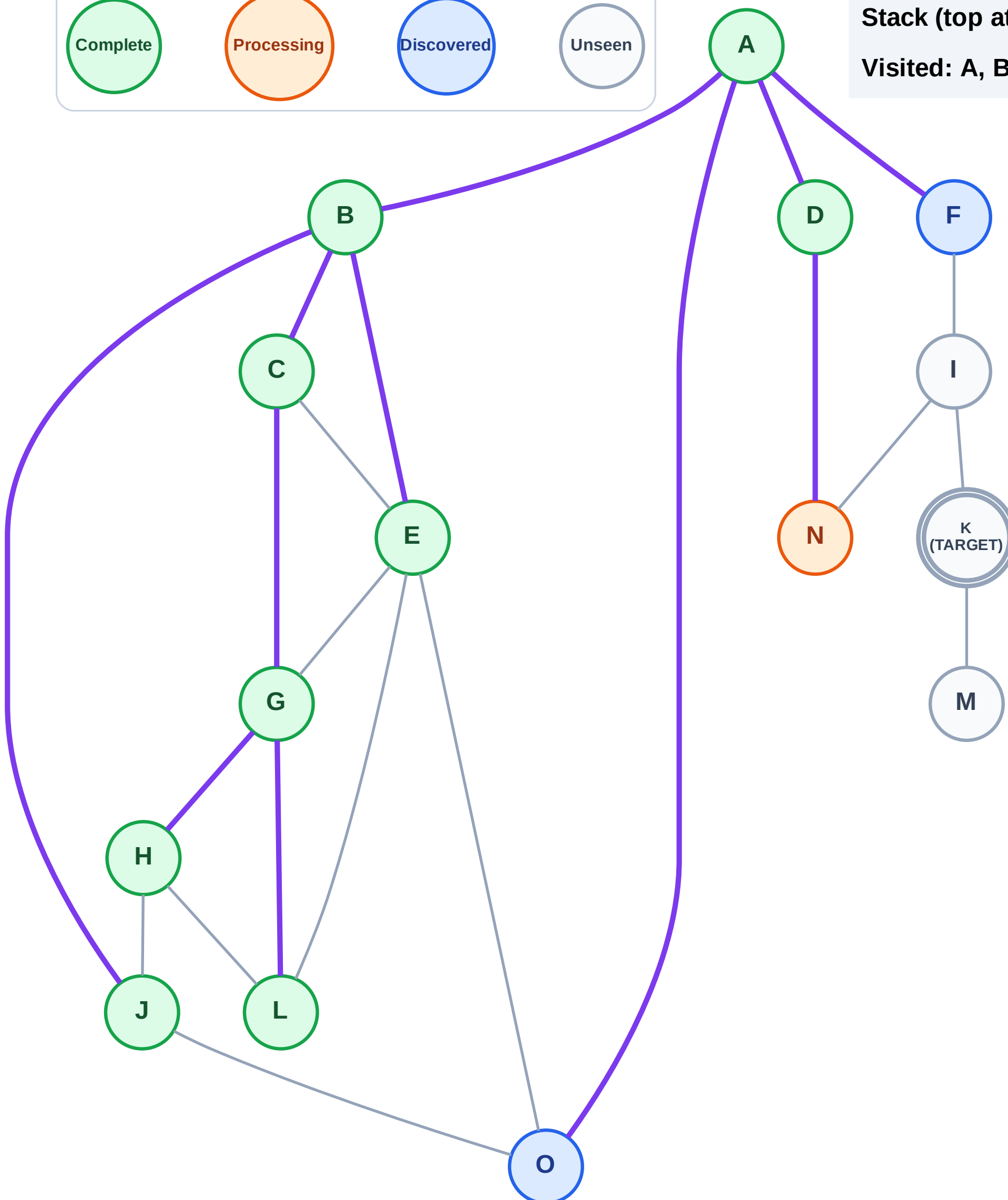


# DFS Traversal

Step 20: Process node N



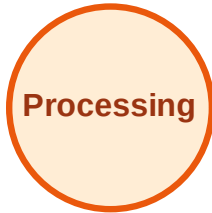
Stack (top at right): O | F  
Visited: A, B, C, D, E, G, H, J, L



# DFS Traversal

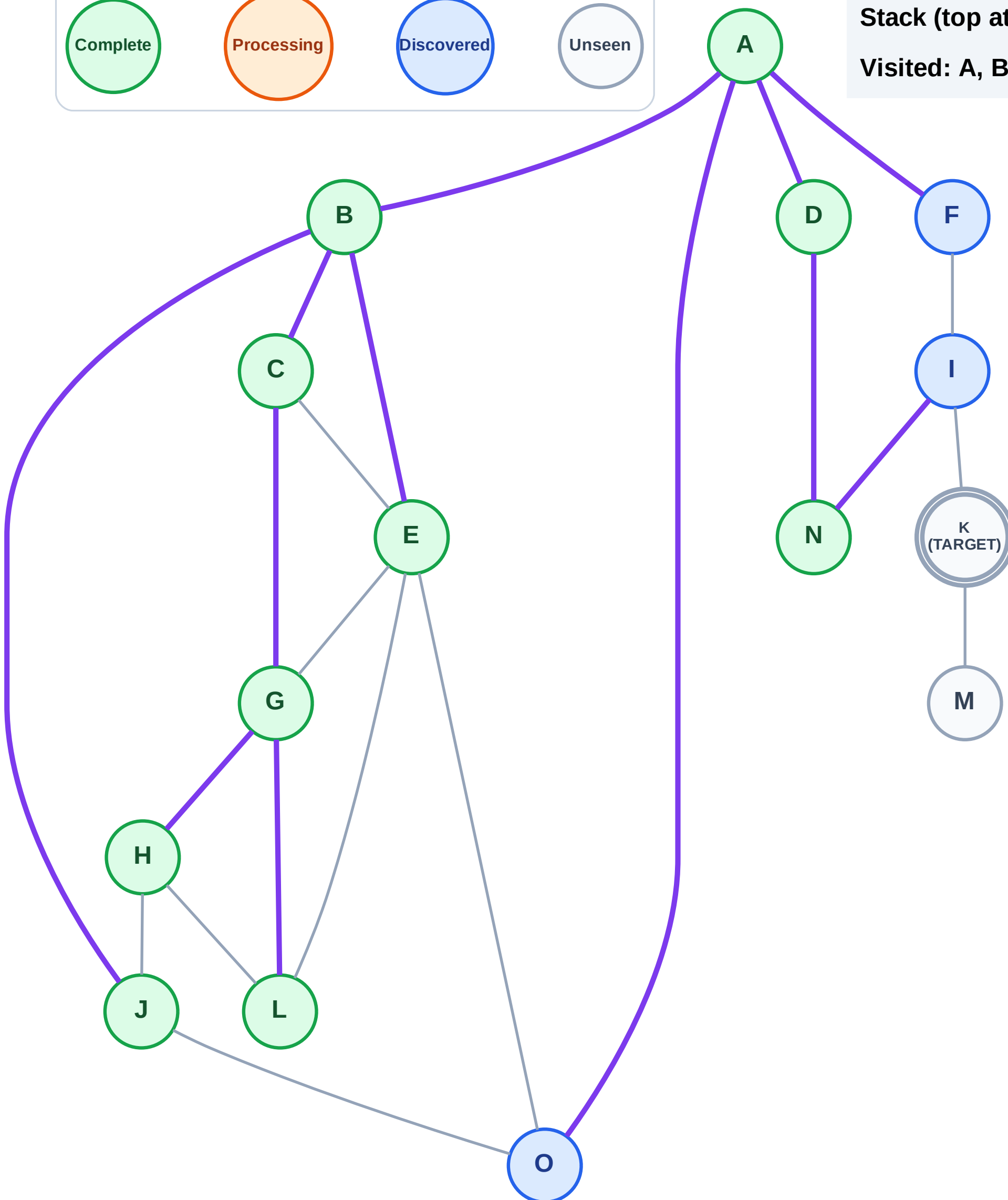
Step 21: Discover I from N

## Legend



Stack (top at right): O | F | I

Visited: A, B, C, D, E, G, H, J, L, N



# DFS Traversal

Step 22: Process node I

## Legend

Complete

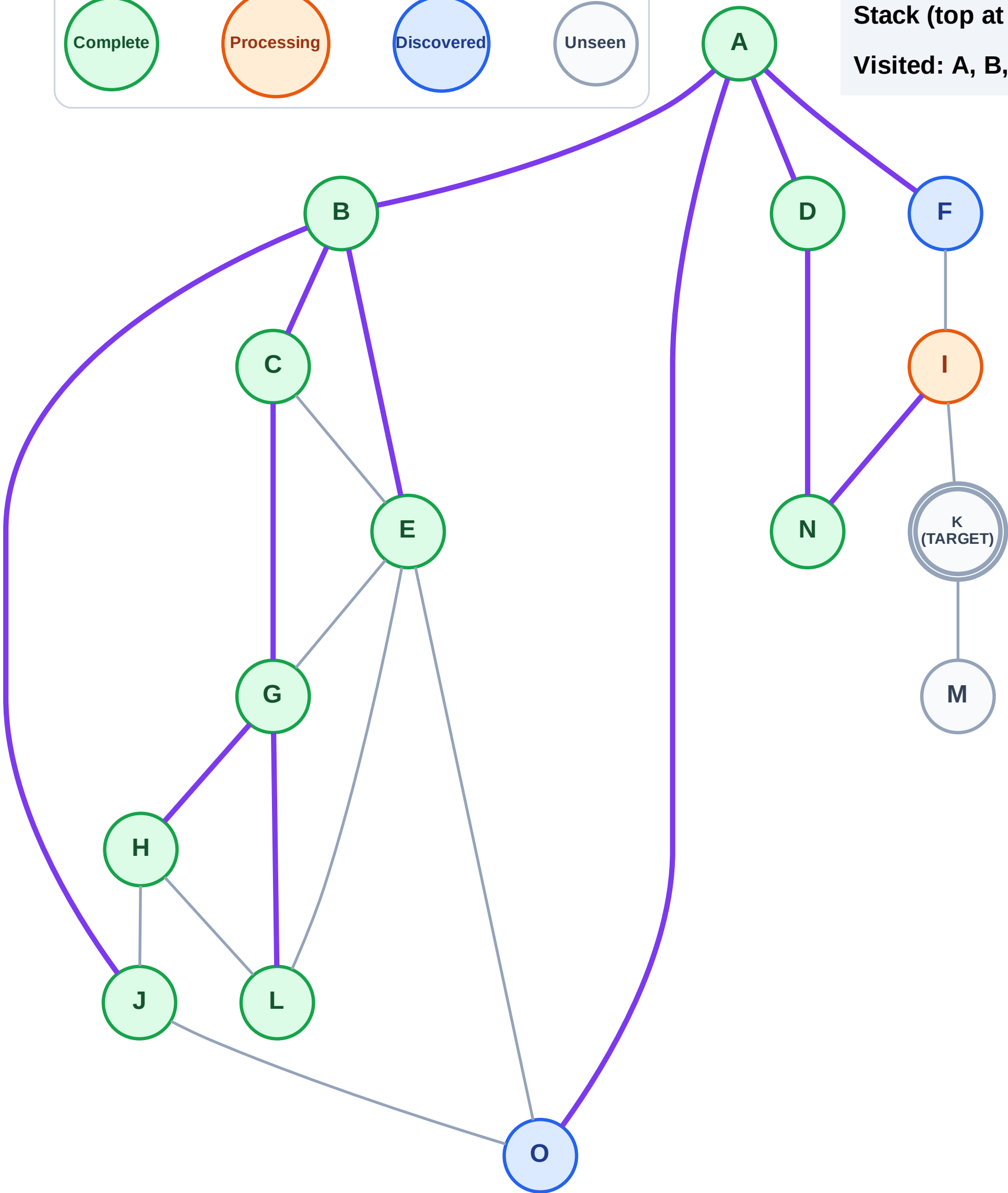
Processing

Discovered

Unseen

Stack (top at right): O | F

Visited: A, B, C, D, E, G, H, J, L, N



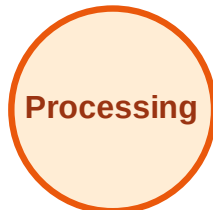
# DFS Traversal

Step 23: Discover K from I

## Legend



Complete



Processing



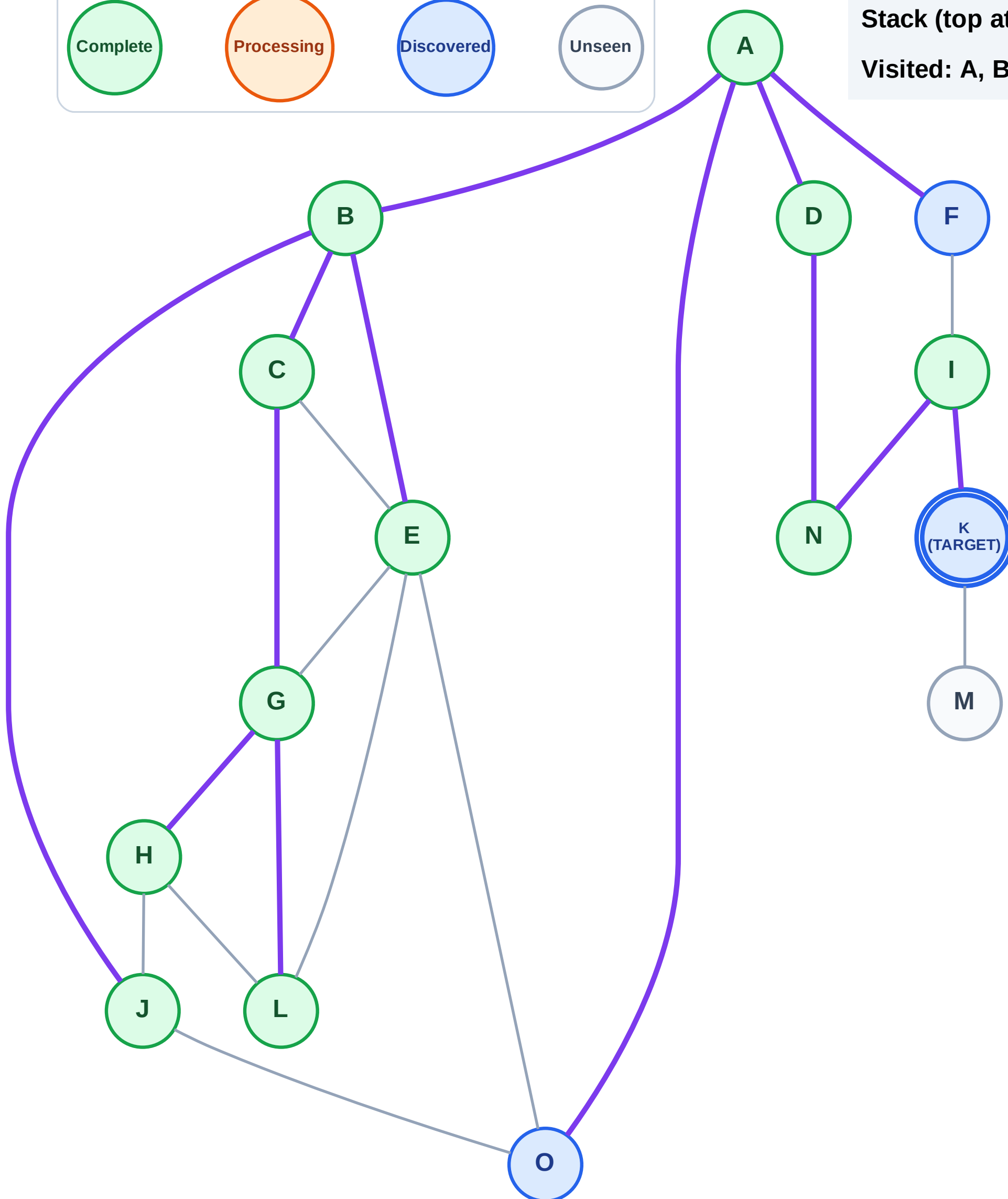
Discovered



Unseen

Stack (top at right): O | F | K

Visited: A, B, C, D, E, G, H, I, J, L, N



# DFS Traversal

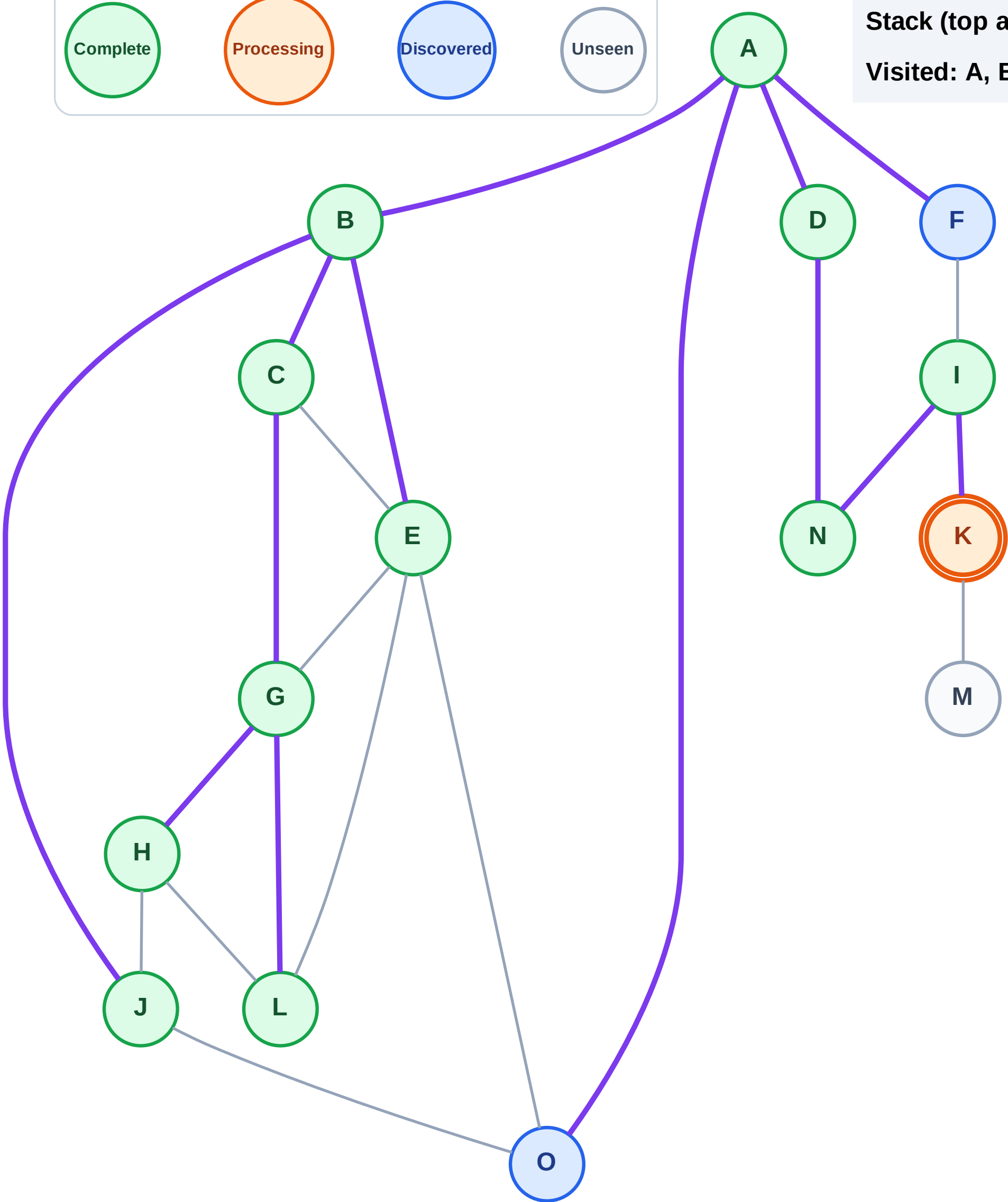
Step 24: Process node K

## Legend



Stack (top at right): O | F

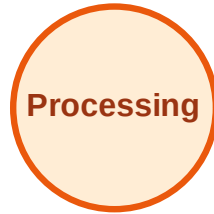
Visited: A, B, C, D, E, G, H, I, J, L, N



# DFS Traversal

Step 25: TARGET NODE K REACHED! Path (4 hops): A → D → N → I → K

## Legend



Stack (top at right): O | F

Visited: A, B, C, D, E, G, H, I, J, K, L, N

